



**DEPARTMENT OF COMPUTER SCIENCE AND
ENGINEERING LAB MANUAL**

CS23432 – Software Construction

(REGULATION 2023)

RAJALAKSHMI ENGINEERING COLLEGE

Thandalam, Chennai-602015

Name:

Register No:

Year / Branch / Section:

Semester:

Academic Year:

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10.		GitHub: Project Structure & Naming Conventions.

EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP
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Aim:

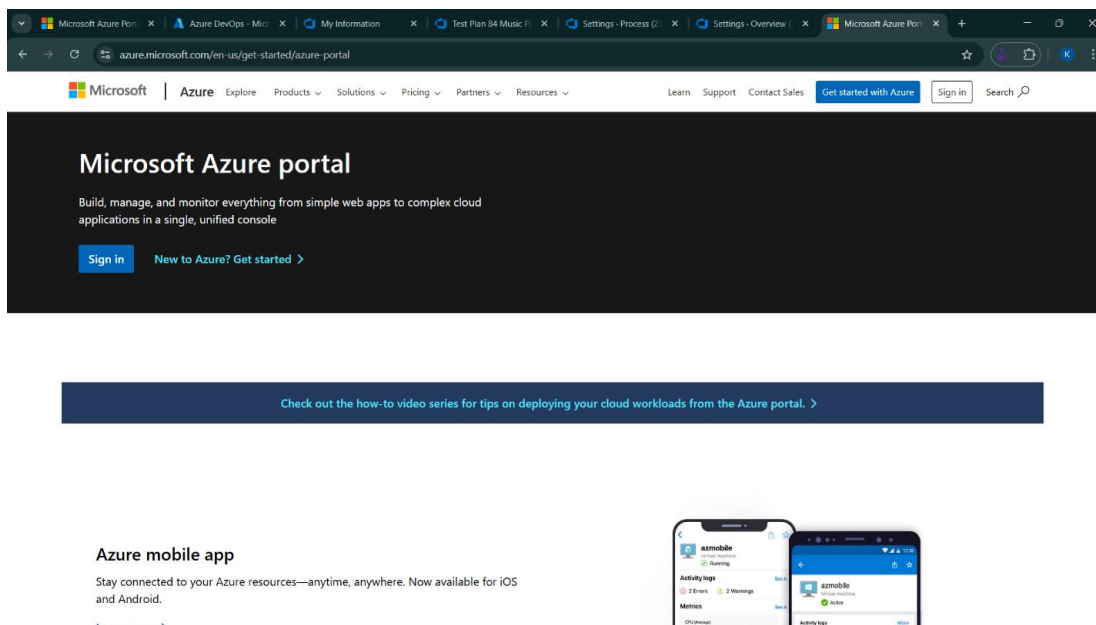
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

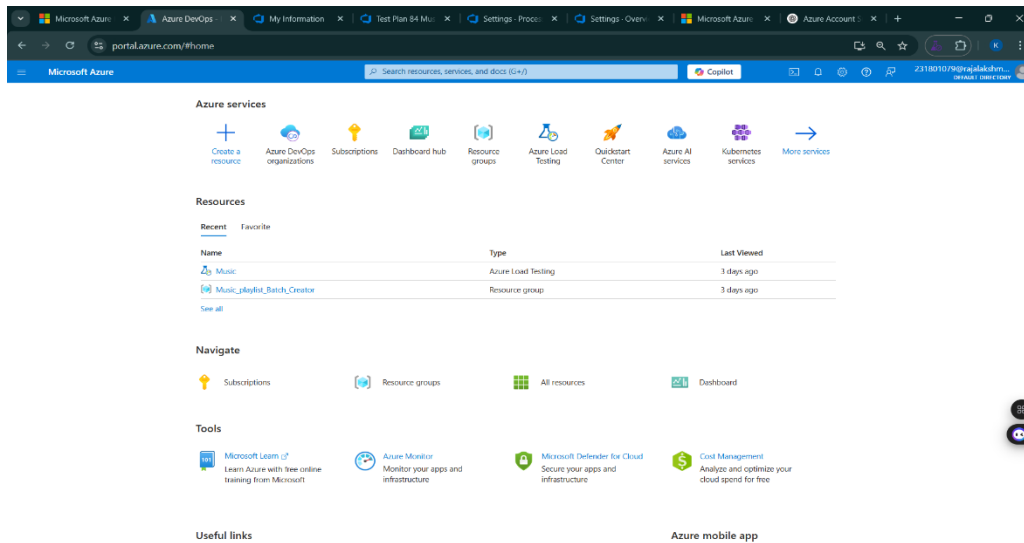
1. Open your web browser and go to the Azure website: <https://azure.microsoft.com/en-us/get-started/azure-portal>.

Sign in using your Microsoft account credentials.

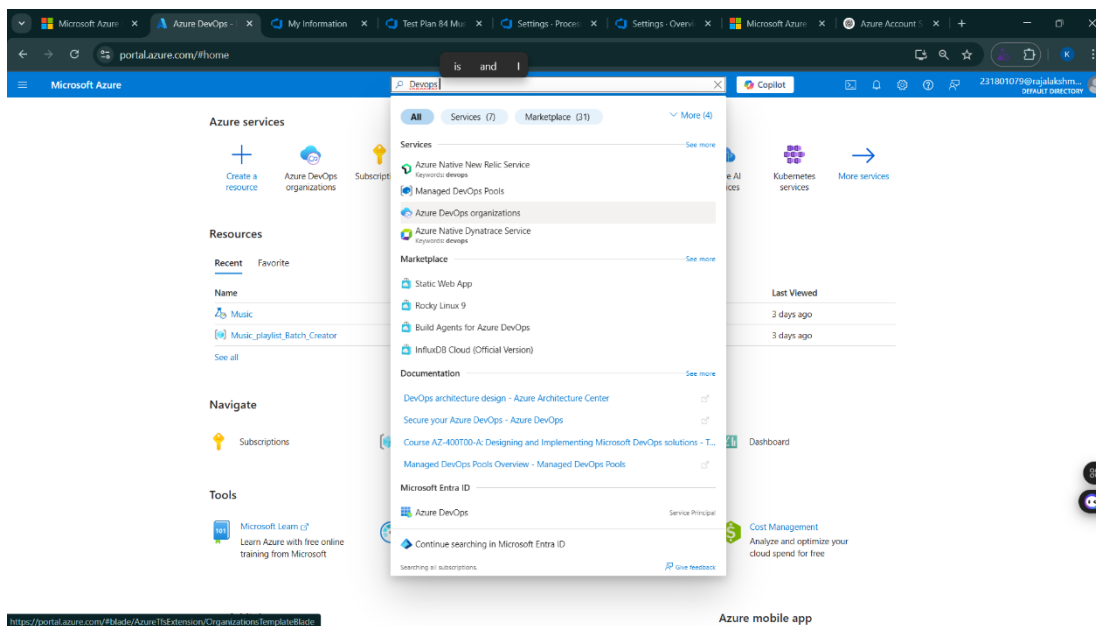
If you don't have a Microsoft account, you can create one here: <https://signup.live.com/?lic=1>



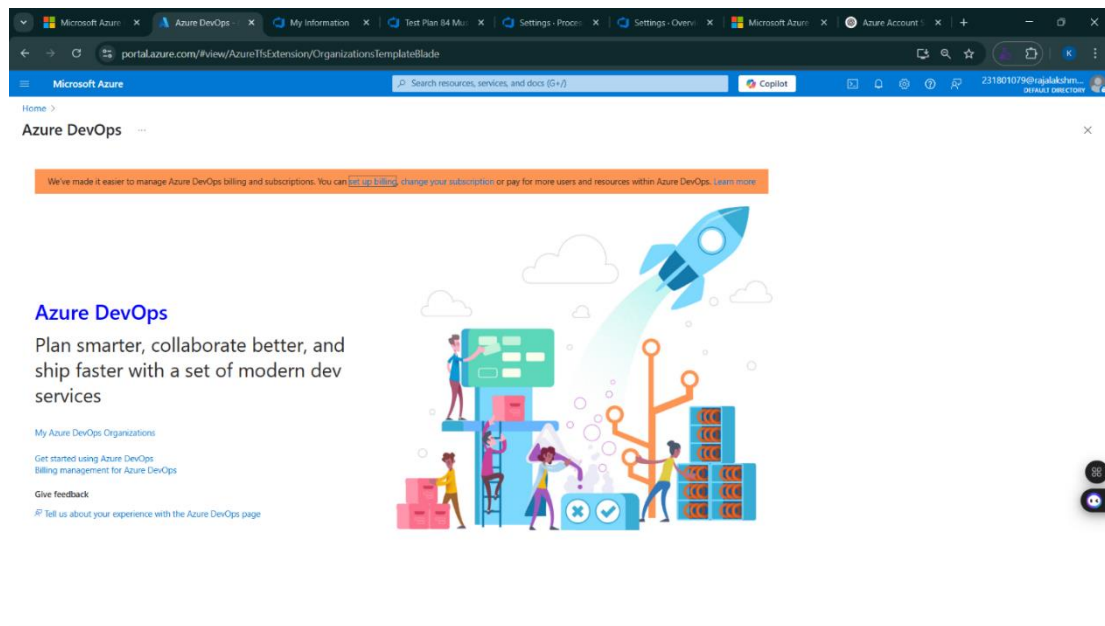
2. Azure home page



3. Open DevOps environment in the Azure platform by typing **Azure DevOps Organizations** in the search bar.



4. Click on the **My Azure DevOps Organization** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

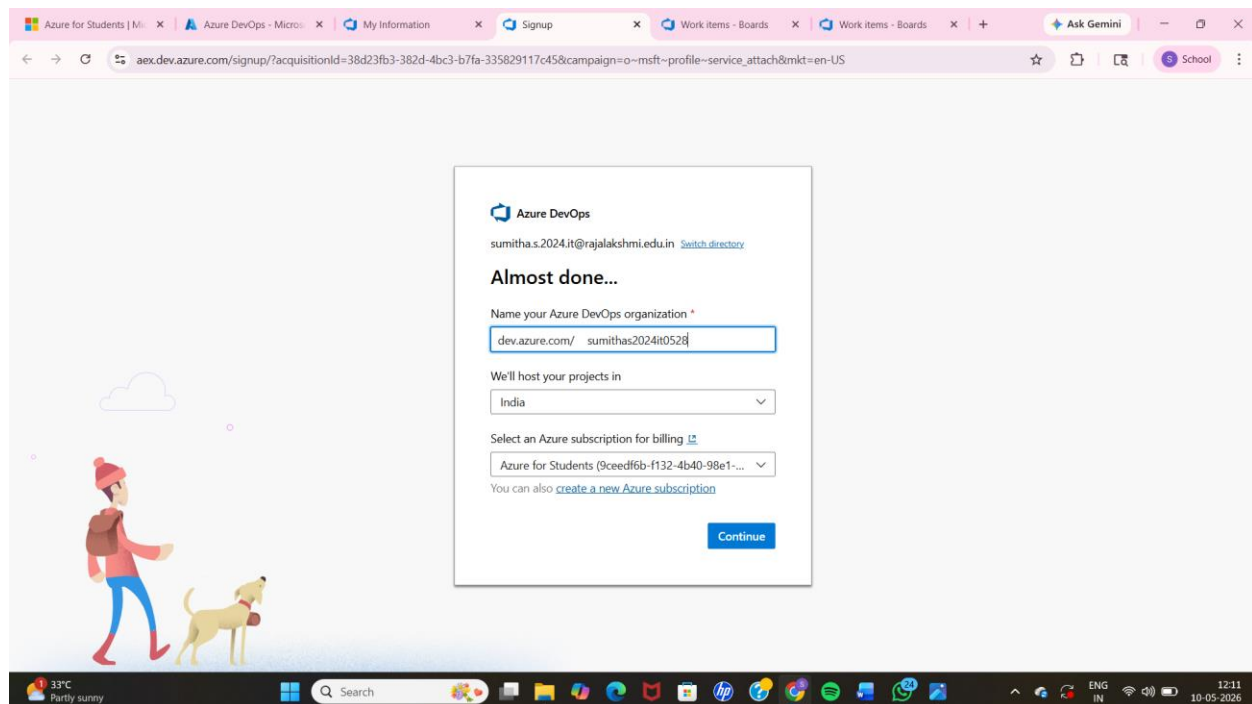
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT
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Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

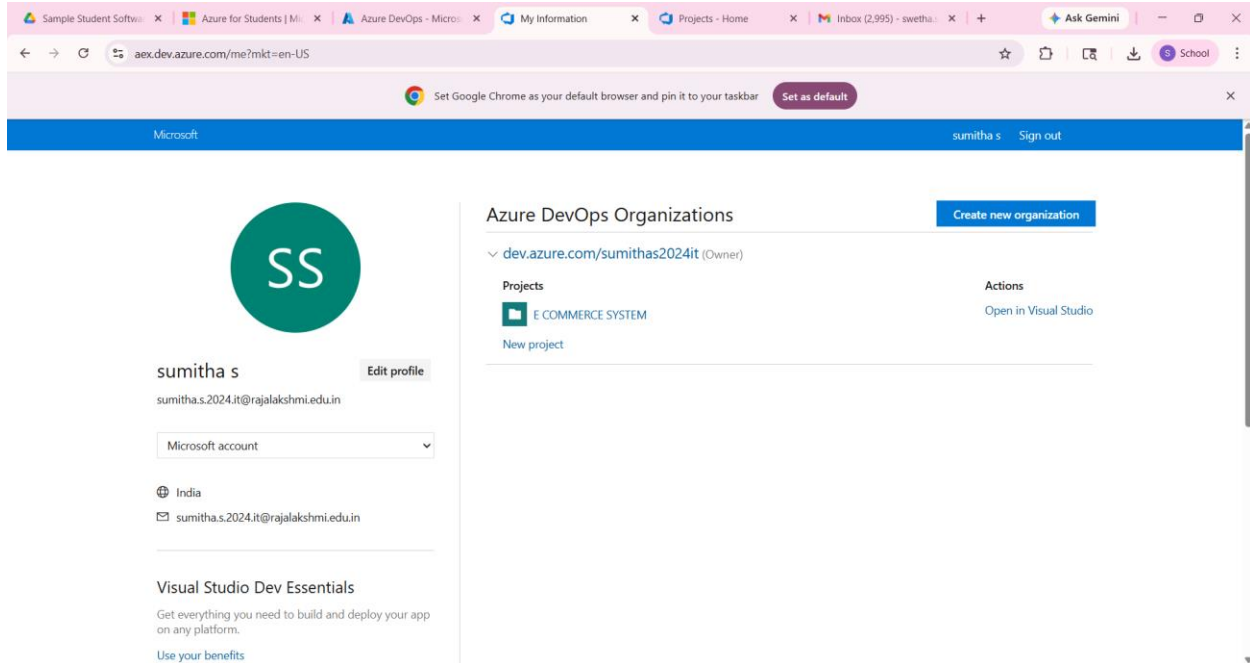
1.Create An Azure Account



2.Create the First Project in Your Organization

- After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.
- On the organization's **Home page**, click on the **New Project** button.
- Enter the project name, description, and visibility options:
 - Name:** Choose a name for the project (e-commerce).
 - Description:** Optionally, add a description to provide more context about the project.
 - Visibility:** Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).
- Once you've filled out the details, click **Create** to set up your first project.

3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



4.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.

Sample Student 5: x Azure for Student: x Azure DevOps - h: x My Information x E COMMERCE SY: x Projects - Home x Inbox (2,995) - s: x + Ask Gemini x

dev.azure.com/sumithas2024it/E%20COMMERCE%20SYSTEM/_backlogs/backlog/E%20COMMERCE%20SYSTEM%20Team/Epics

Set Google Chrome as your default browser and pin it to your taskbar Set as default

sumithas2024it / E COMMERCE SYSTEM / Boards / Backlogs

E COMMERCE SYSTEM Team

+ New Work Item View as Board Column Options

Backlog Analytics

Order	Work Item Type	Title	State	Effort	Busin...	Value Area	Tags
1	Epic	> 🏰 Non functional requirements	⌘ New			Business	
2	Epic	> 🏰 User Stories	⌘ New			Business	

Result:

Successfully created an Azure DevOps project with user story management and agile workflow setup.

SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:
To learn about how to create epics, user story, features, backlogs for your assigned project.

Create Epic, Features, User Stories, Task

The screenshot shows the Azure DevOps interface for the 'E COMMERCE SYSTEM Team'. The 'Backlog' tab is active, displaying a list of 9 user stories. Each story is a 'User Story' type, marked as 'New', and categorized under the 'Business' value area. The iteration paths range from 'E COMMERCE SYSTEM\Iteration 1' to 'E COMMERCE SYSTEM'. The stories are ordered 1 through 9.

Order	Work Item Type	Title	State	Story...	Value Area	Iteration Path	Tags
1	User Story	User Registration	New		Business	E COMMERCE SYSTEM\Iteration 1	
2	User Story	Product Search and Browse	New		Business	E COMMERCE SYSTEM\Iteration 1	
3	User Story	Add to Cart	New		Business	E COMMERCE SYSTEM\Iteration 2	
4	User Story	Checkout and Payment	New		Business	E COMMERCE SYSTEM	
5	User Story	Sell a Product	New		Business	E COMMERCE SYSTEM\Iteration 2	
6	User Story	Order Tracking	New		Business	E COMMERCE SYSTEM\Iteration 2	
7	User Story	Availability & Reliability	New		Business	E COMMERCE SYSTEM	
8	User Story	Scalability	New		Business	E COMMERCE SYSTEM	
9	User Story	Performance	New		Business	E COMMERCE SYSTEM	

1.Fill in Epics

Sample Student Software x Azure for Students | Mi x Azure DevOps - Micro x My Information x E COMMERCE SYSTEM x Projects - Home x + Ask Gemini x

dev.azure.com/sumithas2024it/E%20COMMERCE%20SYSTEM/_backlogs/backlog/E%20COMMERCE%20SYSTEM%20Team/Epics?workitem=16

Set Google Chrome as your default browser and pin it to your taskbar Set as default

W EPIC 16

16 Advanced and reliable e-commerce application to enhance shopping experience

No one selected 0 Comments Add Tag

Save and Close Follow Details 0 0 0

Updated by Sathyaseelan C: Mar 18

Stat: New Area: E COMMERCE SYSTEM Reason: New Iteration: E COMMERCE SYSTEM

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority: 2

Risk

Effort

Business Value

Time Criticality

Start Date: Select a date...

Target Date: Select a date...

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Add an existing work item as a parent

Child

17 Add to Cart Updated 2h ago New

2.Fill in Features

Sample Student Software x Azure for Students | Mi x Azure DevOps - Micro x My Information x E COMMERCE SYSTEM x Projects - Home x + Ask Gemini x

dev.azure.com/sumithas2024it/E%20COMMERCE%20SYSTEM/_backlogs/backlog/E%20COMMERCE%20SYSTEM%20Team/Features?workitem=24

Set Google Chrome as your default browser and pin it to your taskbar Set as default

FEATURE 24

24 view product details

No one selected 0 Comments Add Tag

Save and Close Follow Details 0 0 0

Updated by sumitha s: Just now

Stat: New Area: E COMMERCE SYSTEM Reason: New Iteration: E COMMERCE SYSTEM/Iteration 3

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority: 2

Risk

Effort

Business Value

Time Criticality

Start Date: Select a date...

Target Date: Select a date...

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Add an existing work item as a parent

3.Fill in User Story Details

Sample Student Softw... x Azure for Students | Mi... x Azure DevOps - Micro... x My Information x E COMMERCE SYSTEM x Projects - Home x + Ask Gemini x

dev.azure.com/sumithas2024it/E%20COMMERCE%20SYSTEM/_backlogs/backlog/E%20COMMERCE%20SYSTEM%20Team/Stories?workitem=14

Set Google Chrome as your default browser and pin it to your taskbar Set as default

USER STORY 14

14 User Registration

Sathyaseelan C 0 Comments Add Tag

Save and Close Follow

Updated by sumitha s: 26m ago

Stat: New Area: E COMMERCE SYSTEM Reason: New Iteration: E COMMERCE SYSTEM/Iteration 1

Description

As a new customer, I want to create an account so that I can make purchases and track my orders.

Acceptance Criteria

- User can enter name, email, and password
- Email must be unique and properly formatted
- Password must meet minimum security requirements (e.g., 8 characters)
- Account is created and stored in the database

Planning

Story Points: 5
Priority: 2
Risk:

Classification

Value area: Business

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Add link

Link an Azure Repos [commit](#), [pull request](#) or [branch](#) to see the status of your development. You can also [create a branch](#) to get started.

Related Work

Add link

Parent

16 User Stories

Updated Mar 18 @ New

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

Result:

Thus, the creation of epics, features, user story and task has been created successfully.

SPRINT PLANNING

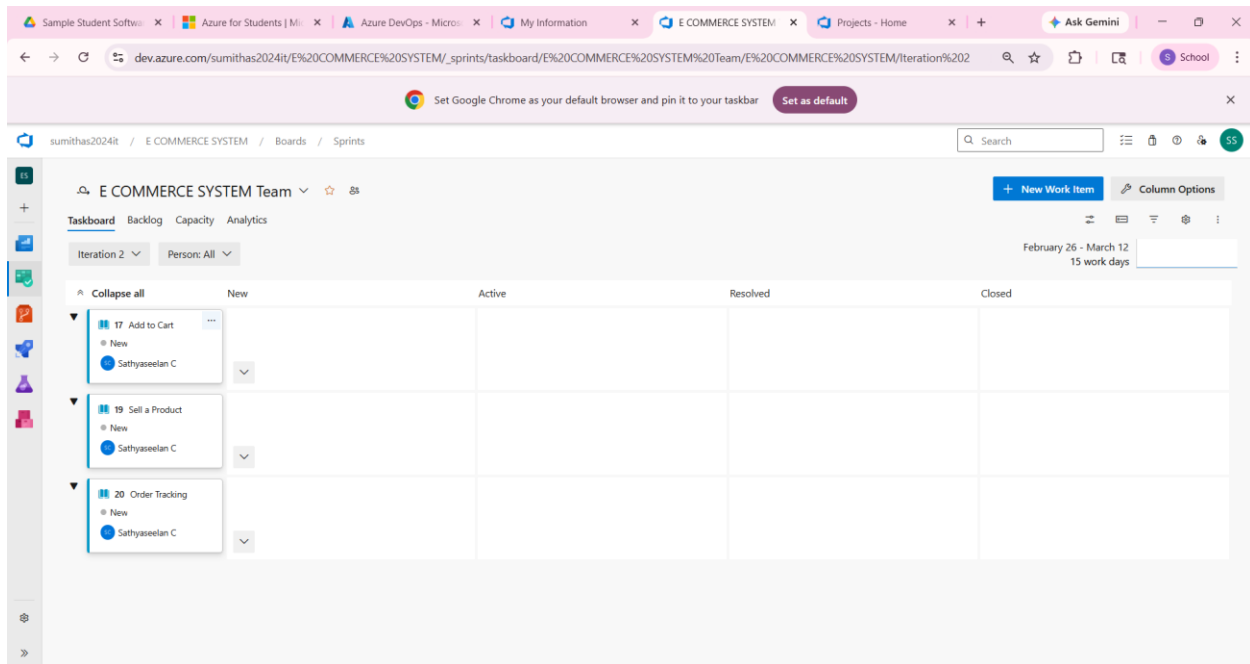
Aim:

To assign user story to specific sprint for the E-commerce Project.

Sprint Planning Sprint 1

The screenshot displays the Azure DevOps Sprints taskboard for the 'E COMMERCE SYSTEM Team'. The interface includes a top navigation bar with various tabs like 'Taskboard', 'Backlog', 'Capacity', and 'Analytics'. The main area shows a Kanban board with columns for 'New', 'Active', 'Resolved', and 'Closed'. Two user stories are visible in the 'New' column: '14 User Registration' and '15 Product Search and Browse', both assigned to 'Sathyaseelan C'. The board is for 'Iteration 1' and covers the period from 'February 10' to 'February 25' (16 work days). A sidebar on the left contains icons for different tools and services.

Sprint 2

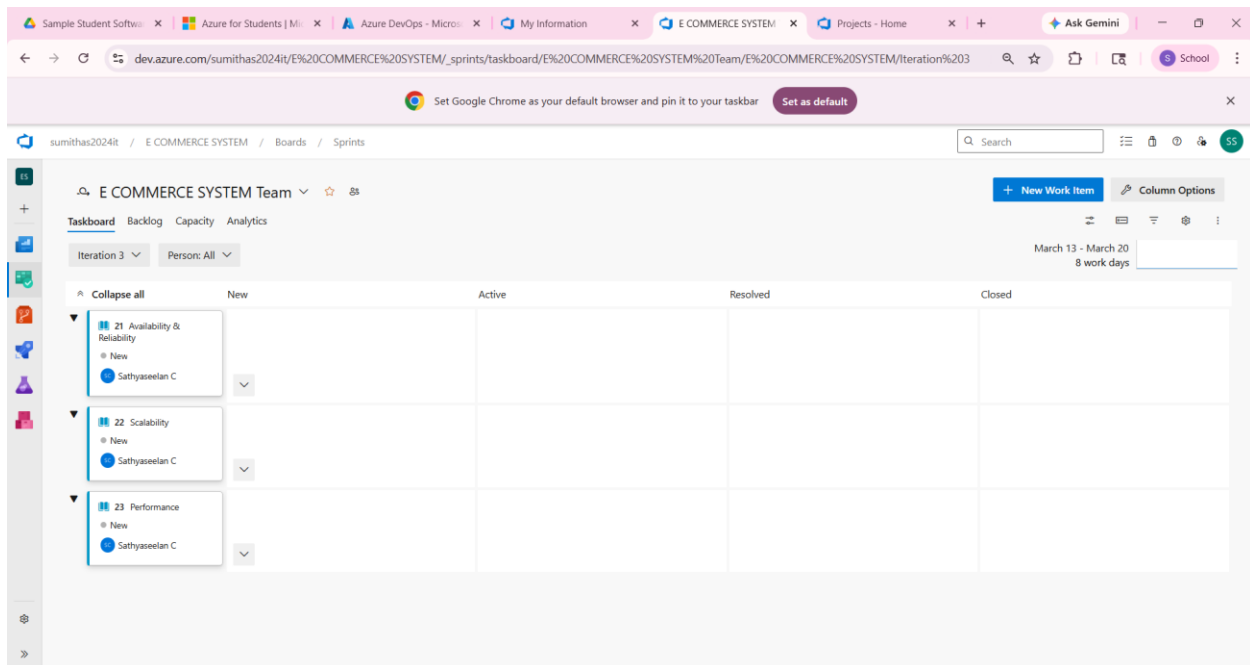


The screenshot shows the Azure DevOps interface for a project named "E COMMERCE SYSTEM". The "Sprints" section is active, displaying "Sprint 2" (Iteration 2) from February 26 to March 12, spanning 15 work days. The taskboard is organized into columns: New, Active, Resolved, and Closed. Three tasks are listed in the "New" column, all assigned to "Sathyaseelan C":

- Task 17: Add to Cart (New)
- Task 19: Sell a Product (New)
- Task 20: Order Tracking (New)

The interface includes a sidebar with navigation icons, a top navigation bar with tabs for "Taskboard", "Backlog", "Capacity", and "Analytics", and a search bar at the top right.

Sprint 3



The screenshot shows the Azure DevOps interface for the same project, "E COMMERCE SYSTEM". The "Sprints" section is active, displaying "Sprint 3" (Iteration 3) from March 13 to March 20, spanning 8 work days. The taskboard is organized into columns: New, Active, Resolved, and Closed. Three tasks are listed in the "New" column, all assigned to "Sathyaseelan C":

- Task 21: Availability & Reliability (New)
- Task 22: Scalability (New)
- Task 23: Performance (New)

The interface includes a sidebar with navigation icons, a top navigation bar with tabs for "Taskboard", "Backlog", "Capacity", and "Analytics", and a search bar at the top right.

Result:

The Sprints are created for the E-commerce Project.

POKER ESTIMATION

Aim:
Create Poker Estimation for the user stories – E-commerce Project.

Poker Estimation

The screenshot shows an Azure DevOps web interface for a work item titled "20 Order Tracking" under the "USER STORY 20" group. The work item is created by Sathyaseelan C. The interface is divided into several sections:

- Description:** "As a customer, I want to track my order so that I know when it will be delivered."
- Acceptance Criteria:**
 - Each order shows status (e.g., placed, shipped, delivered)
 - Estimated delivery date is displayed
 - Status updates in real-time or near real-time
- Planning:**
 - Story Points: 3
 - Priority: 2
 - Risk
- Classification:**
 - Value area: Business
- Deployment:** Includes a link to track releases and turn on deployment status reporting.
- Development:** Includes a link to link an Azure Repos commit, pull request, or branch.
- Related Work:** Includes a link to a parent work item titled "16 Advanced and reliable e-commerce application to enh..." updated 8m ago.

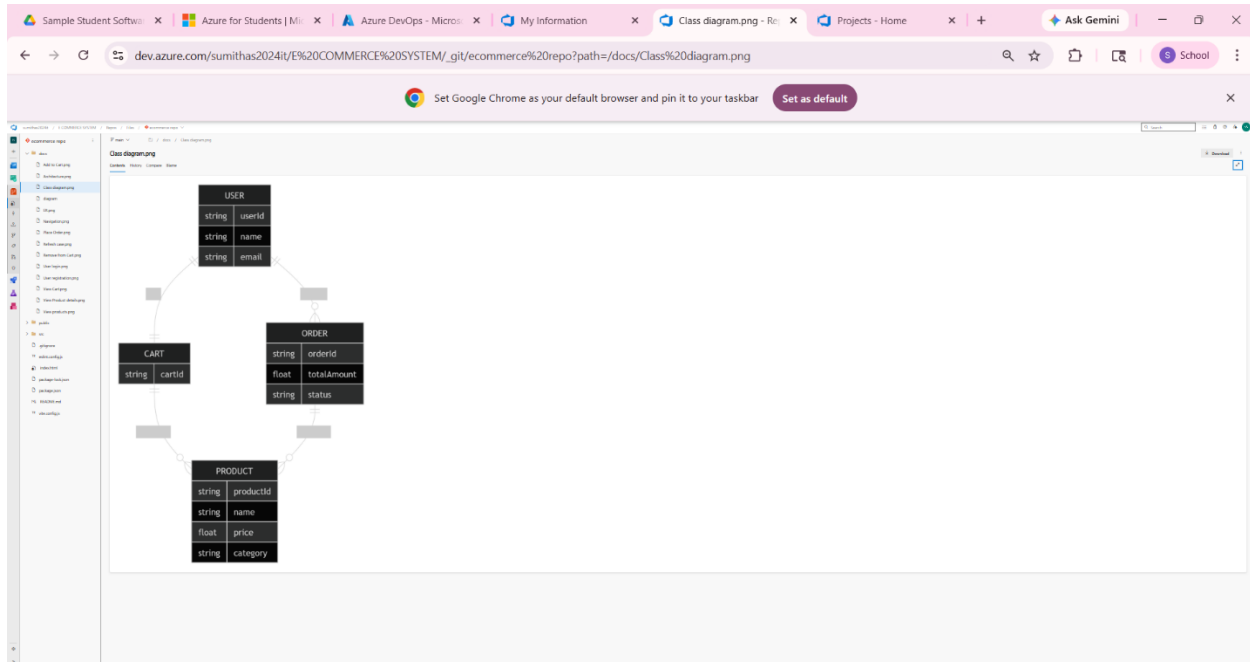
Result:
The Estimation/Story Points is created for the project using Poker Estimation.

DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

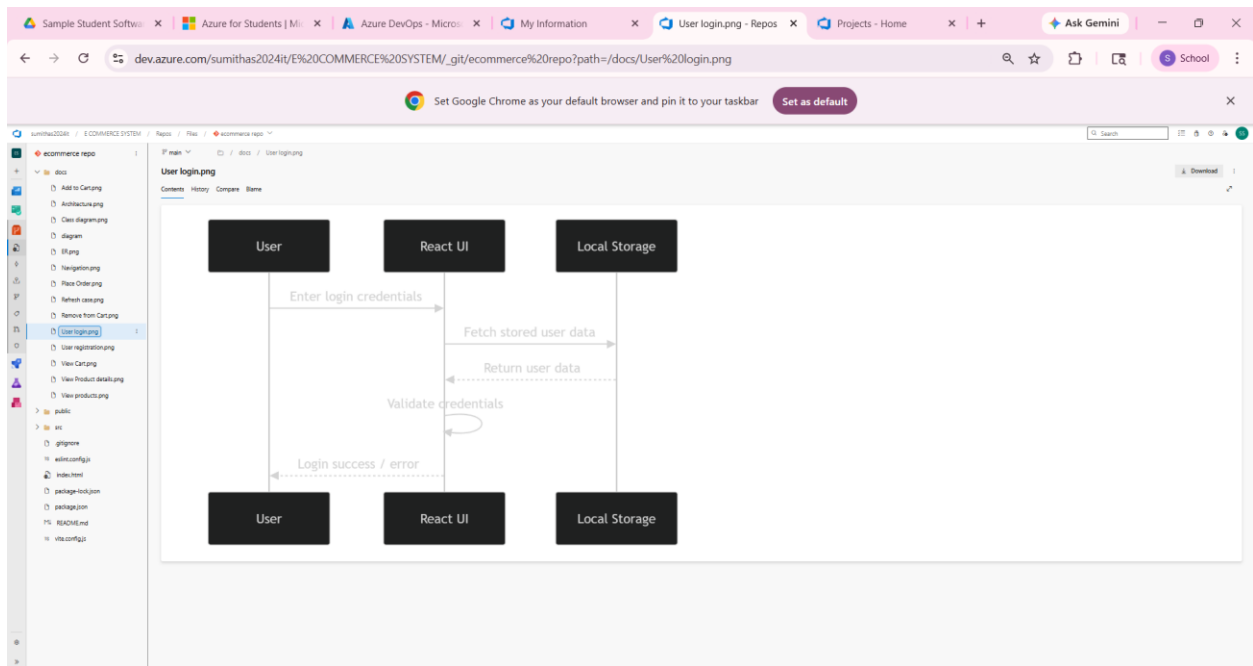
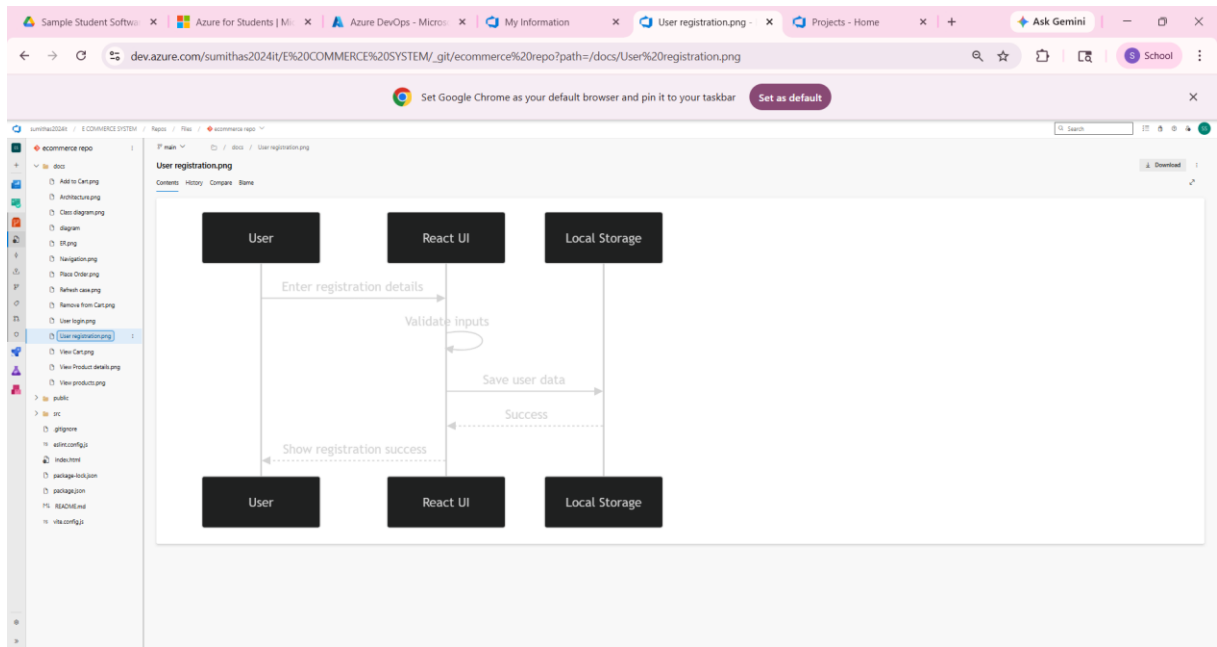
Aim:

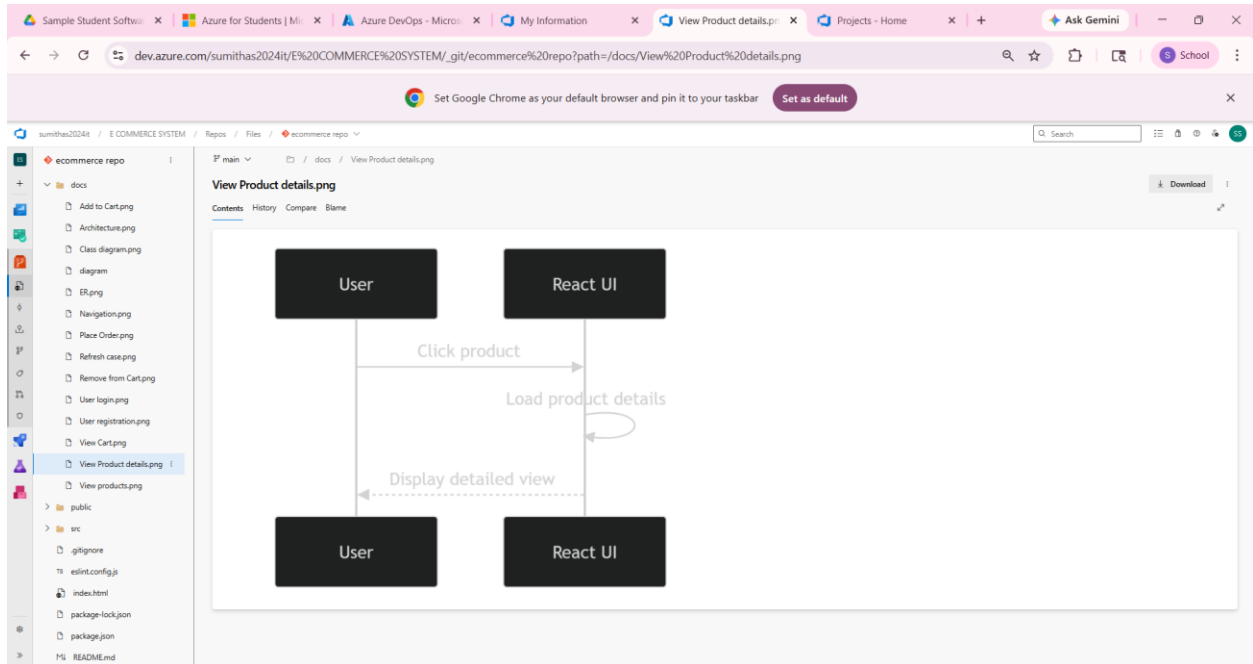
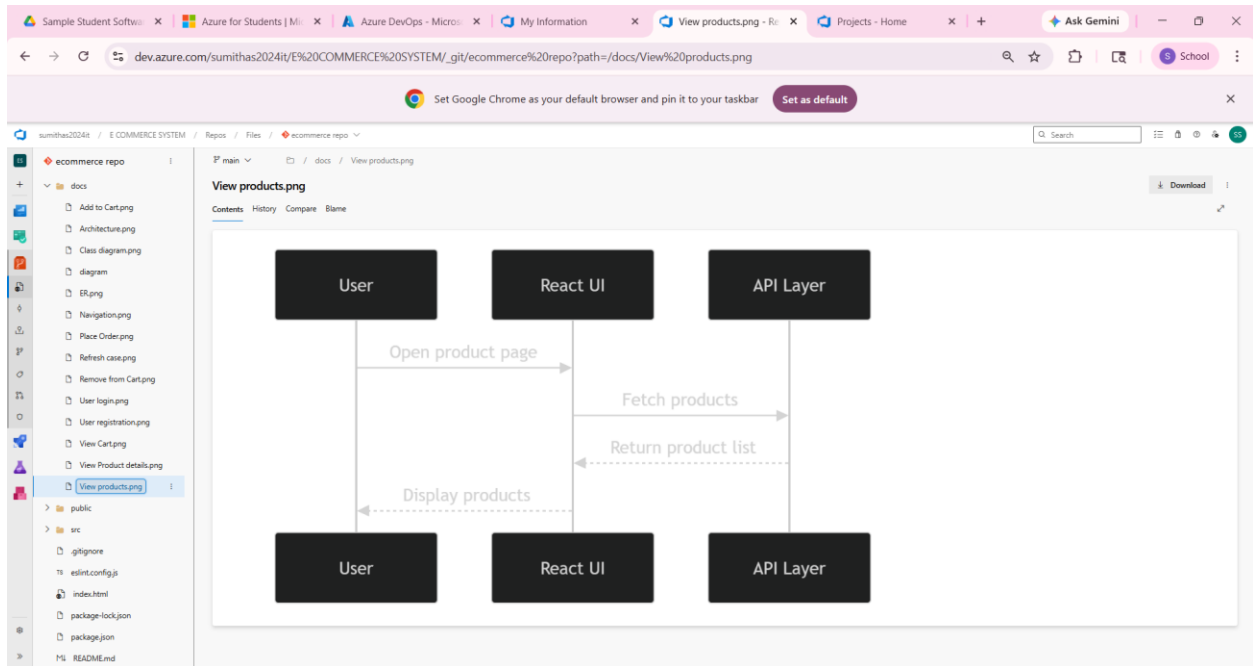
To Design a Class Diagram and Sequence Diagram for the given Project.

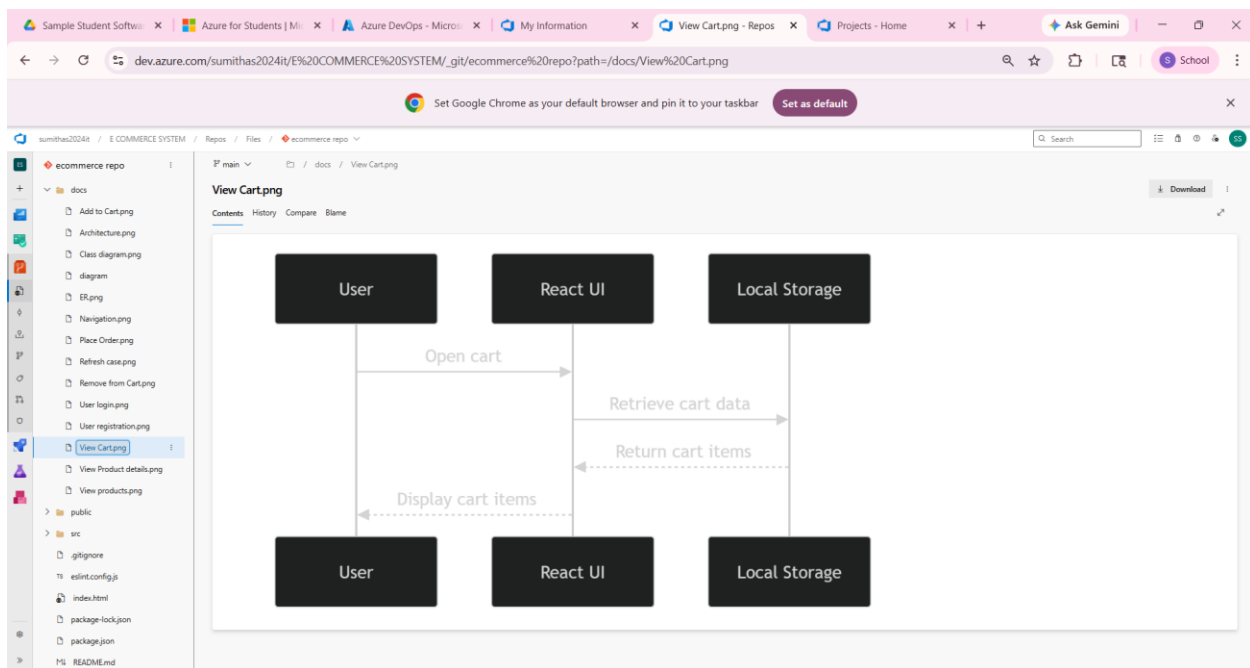
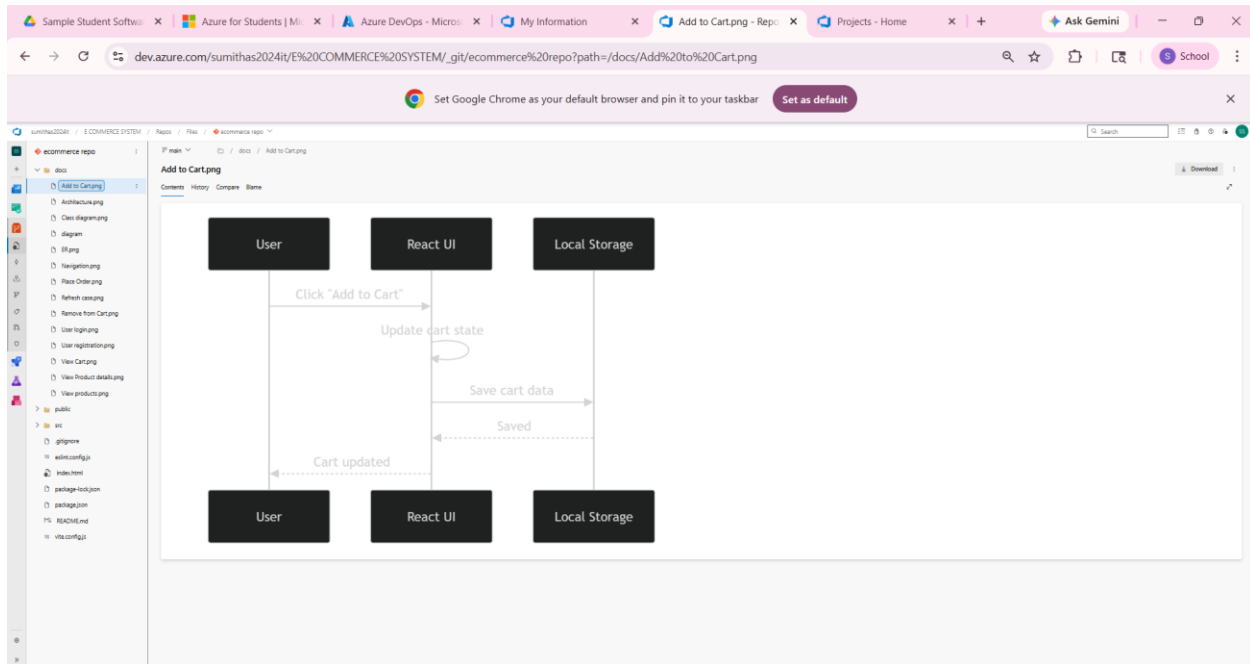
6A. Class Diagram

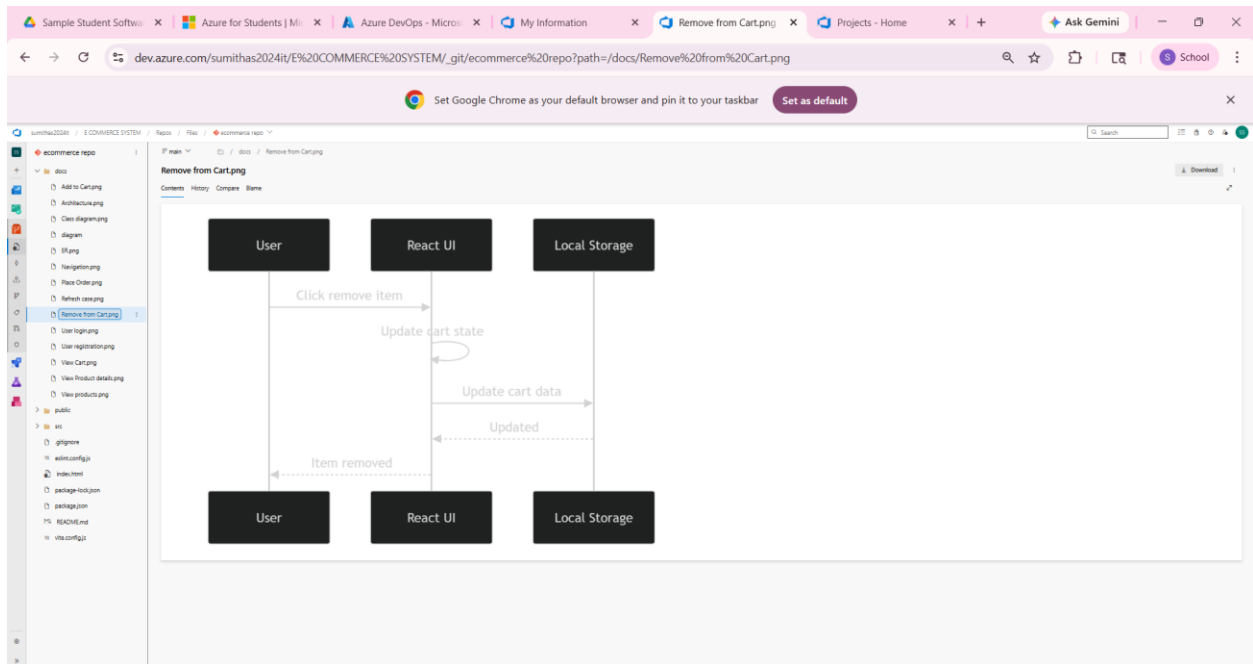
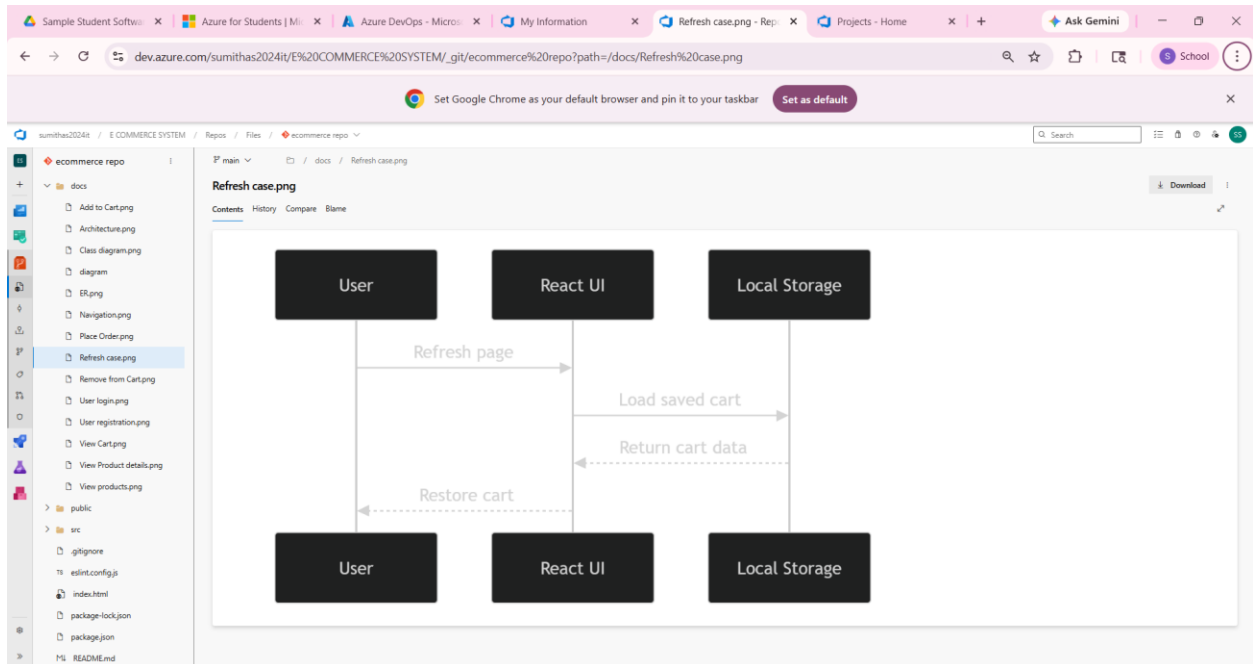


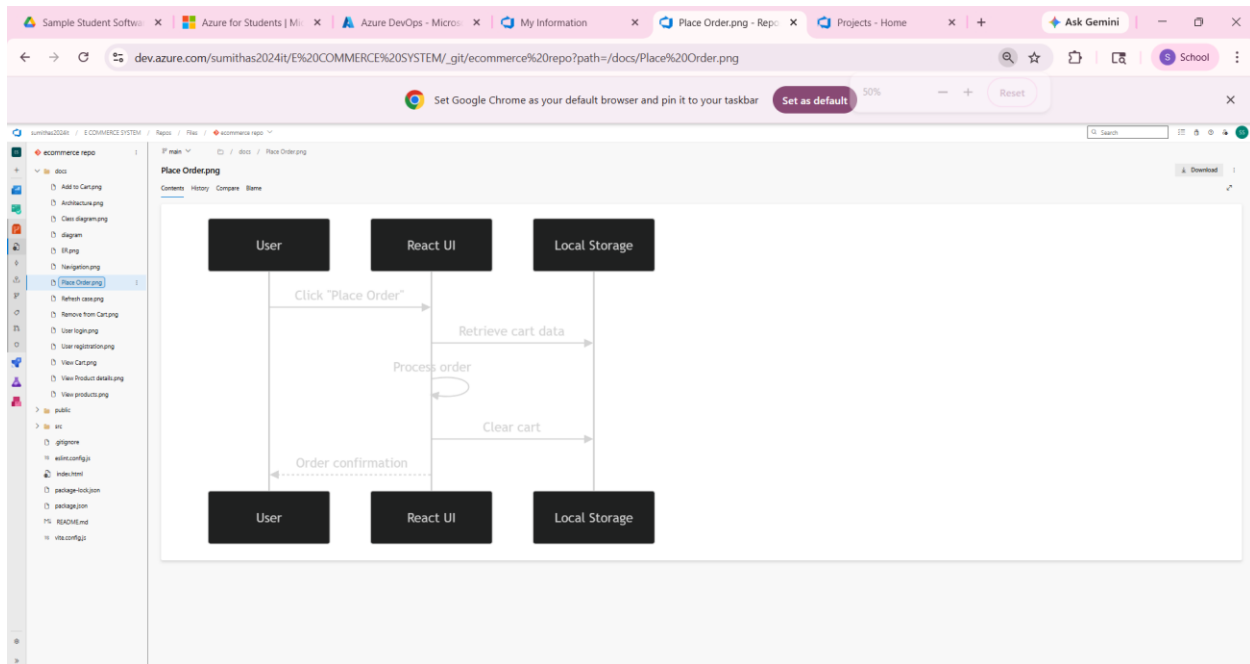
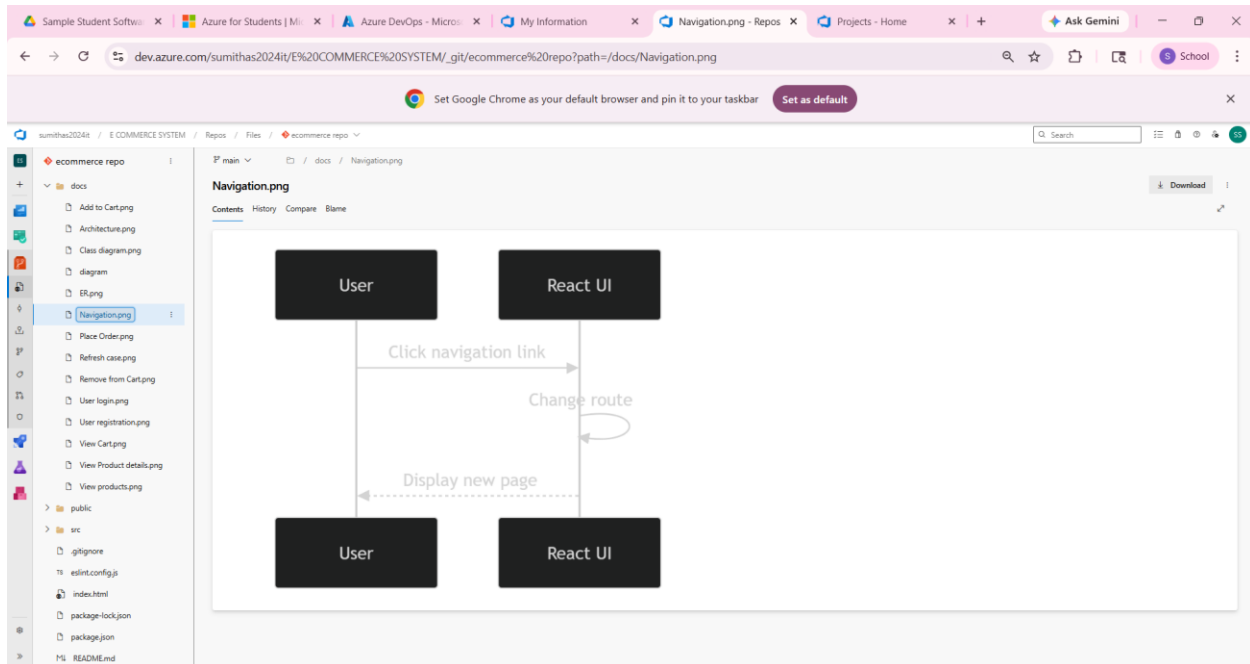
6B. Sequence Diagram











Result:

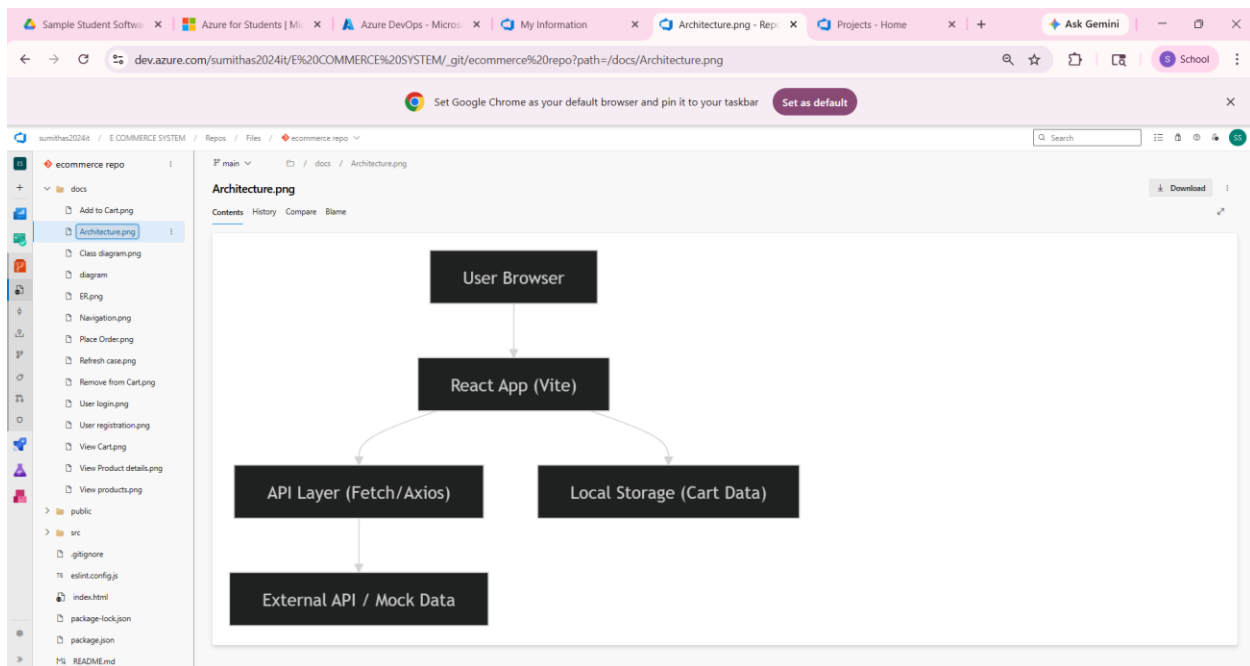
The Class Diagram and Sequence Diagram is designed Successfully for the E-commerce project

DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

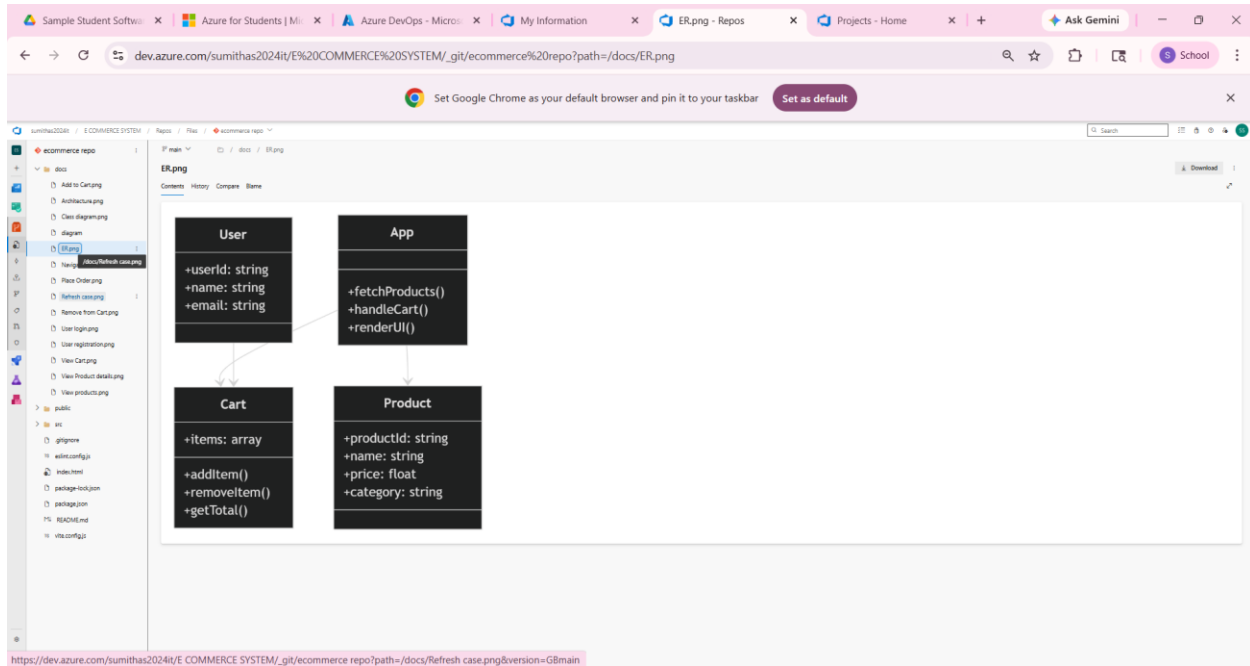
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the E-commerce project.

	TESTING – TEST PLANS AND TEST CASES
--	--

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Test Planning and Test Case

Test Case Design Procedure

1. Understand Core Features of the Application

- User Authentication (Sign Up & Login)
- Product Search
- Add to Cart
- Remove from Cart
- Update Product Quantity
- Place Order and Checkout

2. Define User Interactions

Each test case simulates a real user behavior such as:

- Logging into the application
- Searching for a product
- Adding/removing products from cart
- Updating quantity
- Placing an order

3. Design Happy Path Test Cases

Focused on validating that all features work correctly under normal conditions.

Example:

- User logs in successfully.
- User adds product to cart.
- User places an order successfully.

.

4. Design Error Path Test Cases

Simulate negative or unexpected conditions to verify robustness and error handling.

Example:

- Login with invalid credentials.
- Product not found during search.
- Checkout fails due to payment/network issue.

5. Break Down Steps and Expected Results

Each test case contains:

- Step-by-step actions
- Expected outcomes

This improves clarity for testers and automation scripts.

6. Use Clear Naming and IDs

Examples:

- TC01 – Successful Login
- TC06 – Product Search Failure
- TC10 – Order Placement Failed

This helps in traceability and quick identification.

7. Separate Test Suites

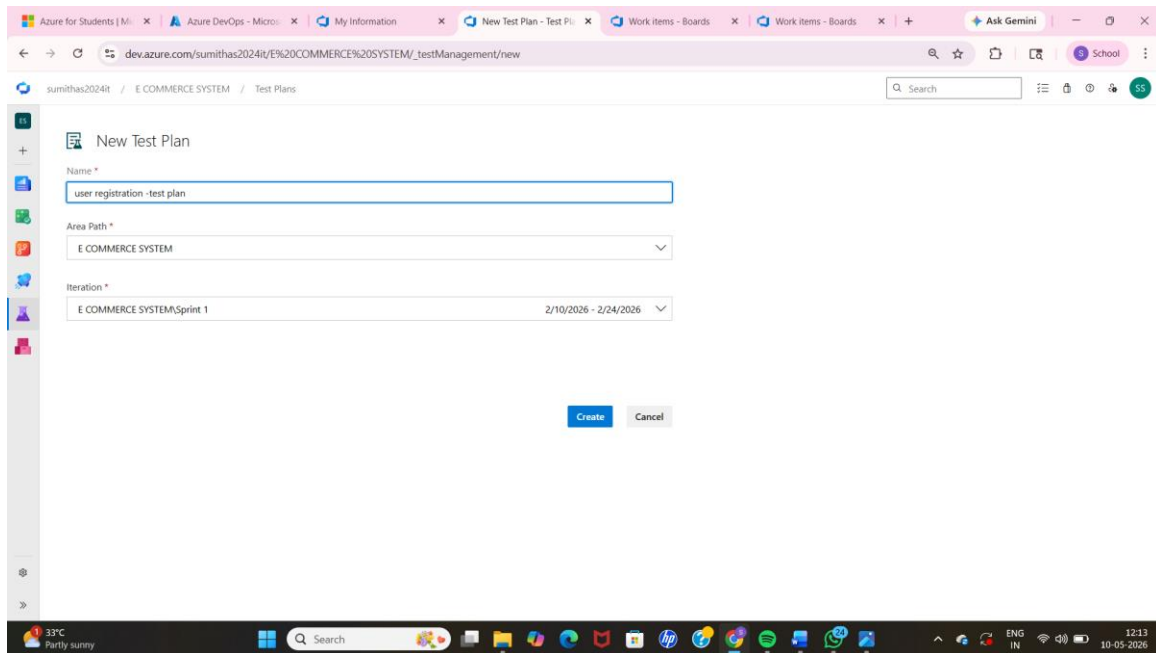
Grouped test cases based on functionality:

- Authentication
- Product Search
- Cart Management
- Checkout and Orders

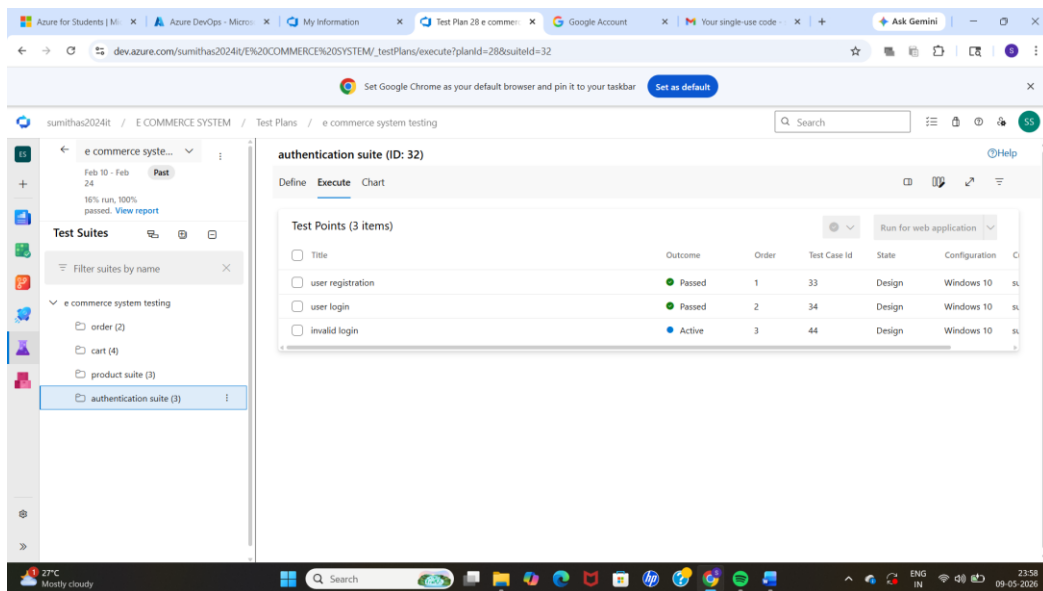
8. Prioritize and Review

- Critical functionalities are marked as high priority.
- Test cases are reviewed for completeness and traceability.

1.New test plan



2. Test suite



3. Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Music Playlist Batch Creator – Test Plans

USER STORIES

- As a user, I want to log in securely so that I can access my account. (ID: 79).
- As a user, I want to search for products easily. (ID: 76).
- As a user, I want to add products to the cart. (ID: 73).
- As a user, I want to remove products from the cart. (ID: 68).
- As a user, I want to update product quantity in the cart. (ID: 65).
- As a user, I want to place an order successfully. (ID: 62).

Test Suites

Test Suit: TS01 – Authentication (ID: 101)

1. TC01 – Successful Login

Action:

- Go to the Login page.
- Enter valid email and password.
- Click “Login”.

Expected Results:

- Login form accepts valid credentials.
- User is redirected to dashboard/homepage.
- Welcome message is displayed.

Type:

Happy Path

2. TC02 – Login with Invalid Password

Action:

- Go to Login page.
- Enter valid email and incorrect password.
- Click “Login”.

Expected Results:

- Error message “Invalid email or password” is displayed.

- User remains on login page.

Type:

Error Path

Test Suit: TS02 – Product Search (ID: 102)

1. TC03 – Search Product Successfully

Action:

- Login successfully.
- Enter product name in search bar.
- Click “Search”.

Expected Results:

- Matching products are displayed.
- Product details such as name, price, and image are shown.

Type:

Happy Path

2. TC04 – Product Search with No Results

Action:

- Login successfully.
- Enter invalid/non-existing product name.
- Click “Search”.

Expected Results:

- Message “No products found” is displayed.
- No product list is shown.

Type:

Error Path

Test Suit: TS03 – Add to Cart (ID: 103)

1. TC05 – Add Product to Cart Successfully

Action:

- Login successfully.
- Search and open a product.
- Click “Add to Cart”.

Expected Results:

- Product is added to cart successfully.
- Cart count increases.
- Success message is displayed.

Type:

Happy Path

2. TC06 – Add Out-of-Stock Product to Cart

Action:

- Login successfully.
- Open an out-of-stock product.
- Click “Add to Cart”.

Expected Results:

- Error message “Product Out of Stock” is displayed.
- Product is not added to cart.

Type:

Error Path

Test Suit: TS04 – Remove from Cart (ID: 104)

1. TC07 – Remove Product from Cart Successfully

Action:

- Login successfully.
- Navigate to Cart page.
- Click “Remove” on a product.

Expected Results:

- Product is removed from cart.
- Cart total updates correctly.

Type:

Happy Path

2. TC08 – Remove Product Failure

Action:

- Login successfully.
- Navigate to Cart page.
- Disconnect internet or simulate server issue.
- Click “Remove”.

Expected Results:

- Error message “Unable to remove item” is displayed.
- Product remains in cart.

Type:

Error Path

Test Suit: TS05 – Update Quantity (ID: 105)**1. TC09 – Update Product Quantity Successfully****Action:**

- Login successfully.
- Navigate to Cart page.
- Increase product quantity.
- Click “Update”.

Expected Results:

- Product quantity is updated successfully.
- Total price updates automatically.

Type:

Happy Path

2. TC10 – Update Quantity with Invalid Value**Action:**

- Login successfully.
- Navigate to Cart page.
- Enter quantity as 0 or negative number.
- Click “Update”.

Expected Results:

- Error message “Invalid quantity” is displayed.
- Quantity remains unchanged.

Type:

Error Path

Test Suit: TS06 – Place Order (ID: 106)

1. TC11 – Place Order Successfully

Action:

- Login successfully.
- Add products to cart.
- Proceed to Checkout.
- Enter valid shipping and payment details.
- Click “Place Order”.

Expected Results:

- Order is placed successfully.
- Order confirmation message is displayed.
- Order ID is generated.

Type:

Happy Path

2. TC12 – Place Order Failure Due to Payment Error

Action:

- Login successfully.
- Add products to cart.
- Proceed to Checkout.
- Enter invalid payment details.
- Click “Place Order”.

Expected Results:

- Error message “Payment Failed” is displayed.
- Order is not placed.

Type:

Error Path

Test Cases

TEST CASE 33

33 user registration

sumitha s

0 Comments Add Tag

Save and Close Follow

State Design Area E COMMERCE SYSTEM Reason New Iteration E COMMERCE SYSTEM\Sprint 1 Updated by sumitha s Apr 28

Steps Summary Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	open registration page		
2.	enter username		
3.	enter email		
4.	enter password		
5.	click register	account created successfully	

Click or type here to add a step

Parameter values

Recent test results

- Passed by with Windows 10 Completed 6:21 PM, 4/28/2026
- Passed by with Windows 10 Completed 6:20 PM, 4/28/2026

View all test result history

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

Development

Related Work

TEST CASE 46

46 place order

sumitha s

0 Comments Add Tag

Save and Close Follow

State Design Area E COMMERCE SYSTEM Reason New Iteration E COMMERCE SYSTEM\Sprint 1 Updated by sumitha s Apr 28

Steps Summary Associated Automation

Steps

Steps	Action	Expected result	Attachments
1.	add item to cart		
2.	checkout		
3.	confirm order		

Click or type here to add a step

Parameter values

Recent test results

Run manual tests with web runner. [Learn how to get started](#)

Deployment

To track releases associated with this work item, go to [Releases](#) and turn on deployment status reporting for Boards in your pipeline's Options menu. [Learn more about deployment status reporting](#)

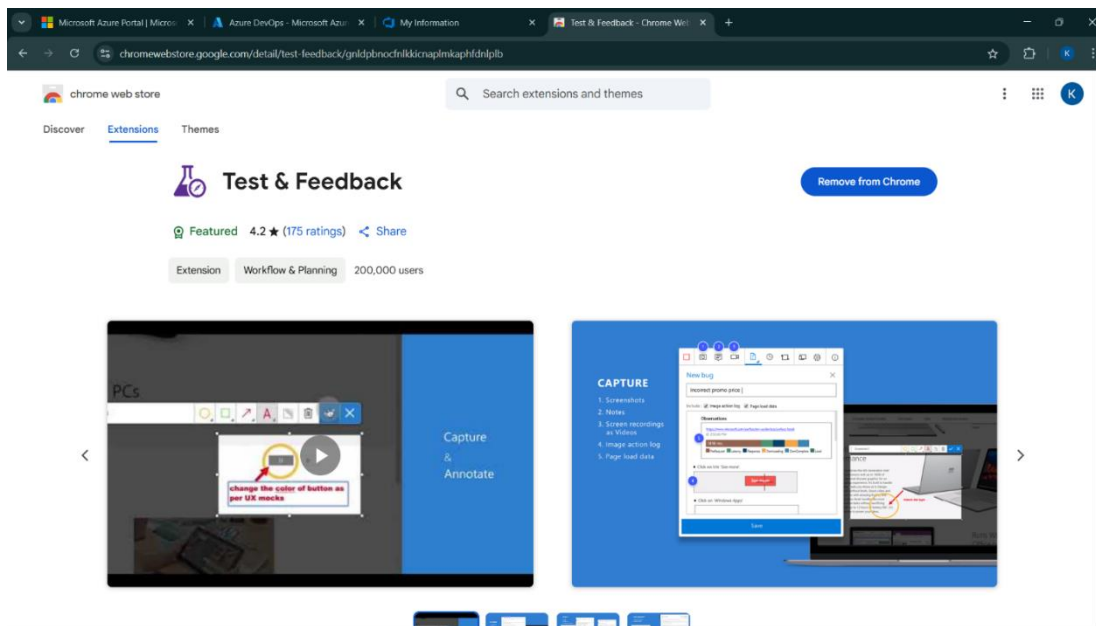
Development

Related Work

Add link

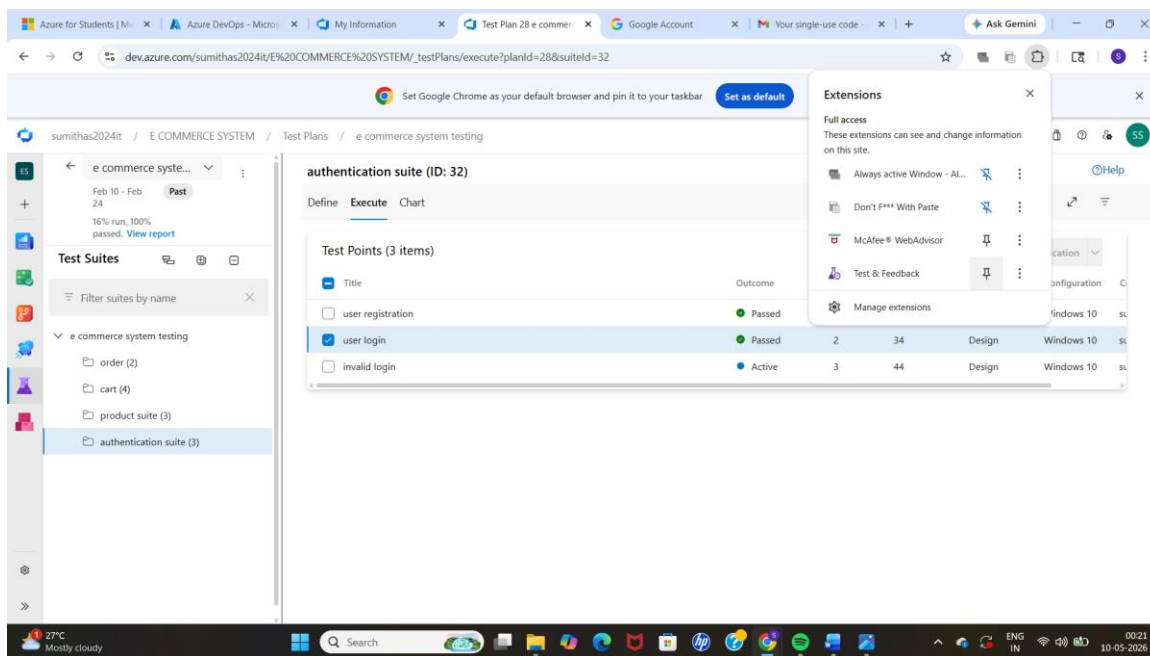
[Add an existing work item as a parent](#)

4.Installation of test



Test and feedback

Showing it as an extension



5. Running the test cases

dev.azure.com/sumithas2024it/E%20COMMERCE%20SYSTEM/_testPlans/execute?planId=288&suiteId=32

Set Google Chrome as your default browser and pin it to your taskbar

sumithas2024it / E COMMERCE SYSTEM / Test Plans / e commerce system testing

authentication suite (ID: 32)

Define Execute Chart

Test Suites

Filter suites by name

e commerce system testing

order (2)

cart (4)

product suite (3)

authentication suite (3)

Test Points (3 items)

Title	Outcome	Order	Test Case Id	State	Configuration
☐ user registration	Passed	1	33	Design	Windows 10
☒ user login	Passed	2	34	Design	Windows 10
☐ invalid login	Active	3	44	Design	Windows 10

View execution history

Mark Outcome

Run

Reset test to active

Edit test case

Assign tester

View test result

Run for web application

Run for desktop application

Run with options

27°C Mostly cloudy

Search

ENG IN

00:13 10-05-2026

Runner - Test Plans - Google Chrome

dev.azure.com/sumithas2024it/E%20COMMERCE%20SYSTEM/_testExecution/Index

Save and close Create bug

34: user login

- open login page
- enter valid username/password
- click login

Hot days ahead 27°C

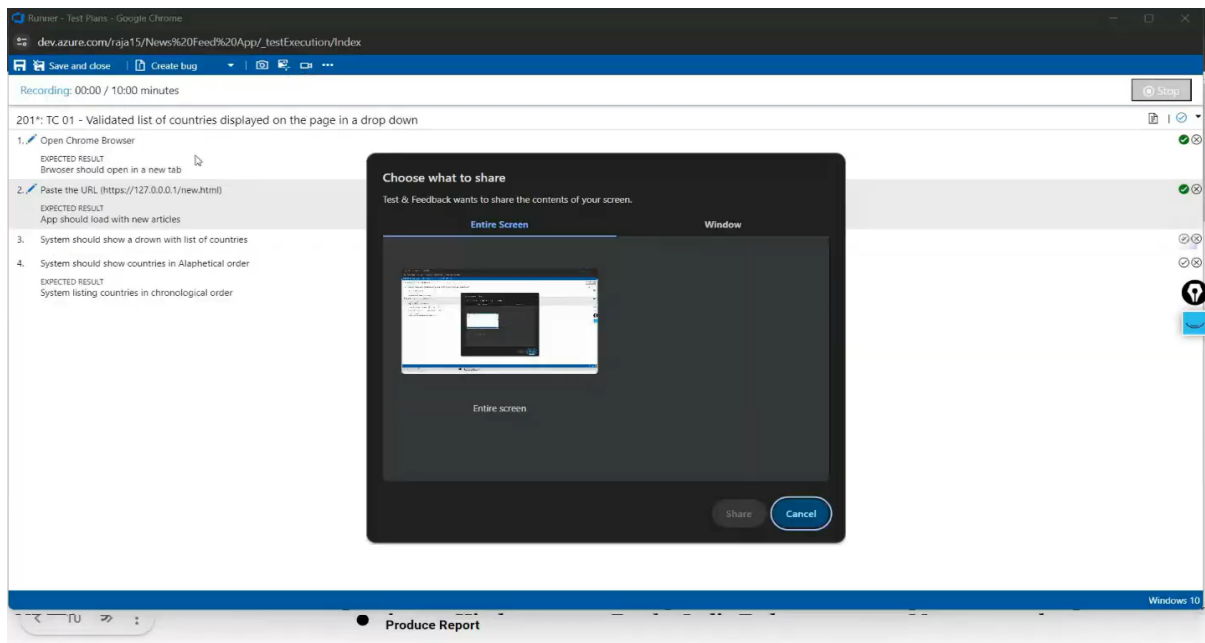
Search

ENG IN

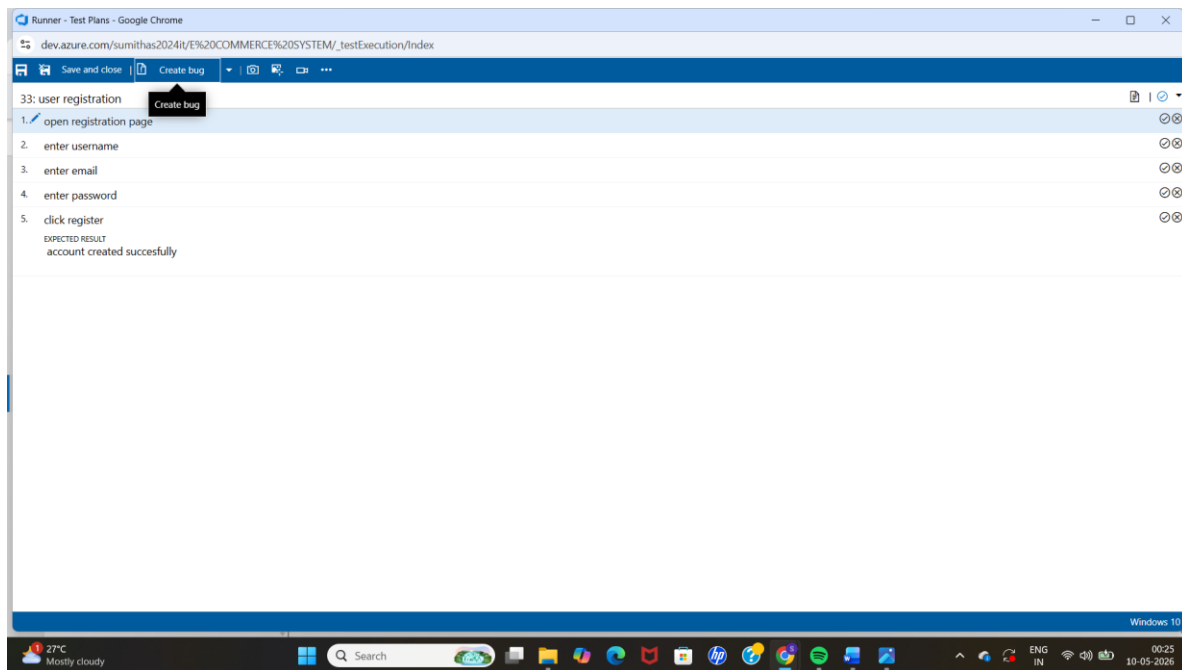
00:15 10-05-2026

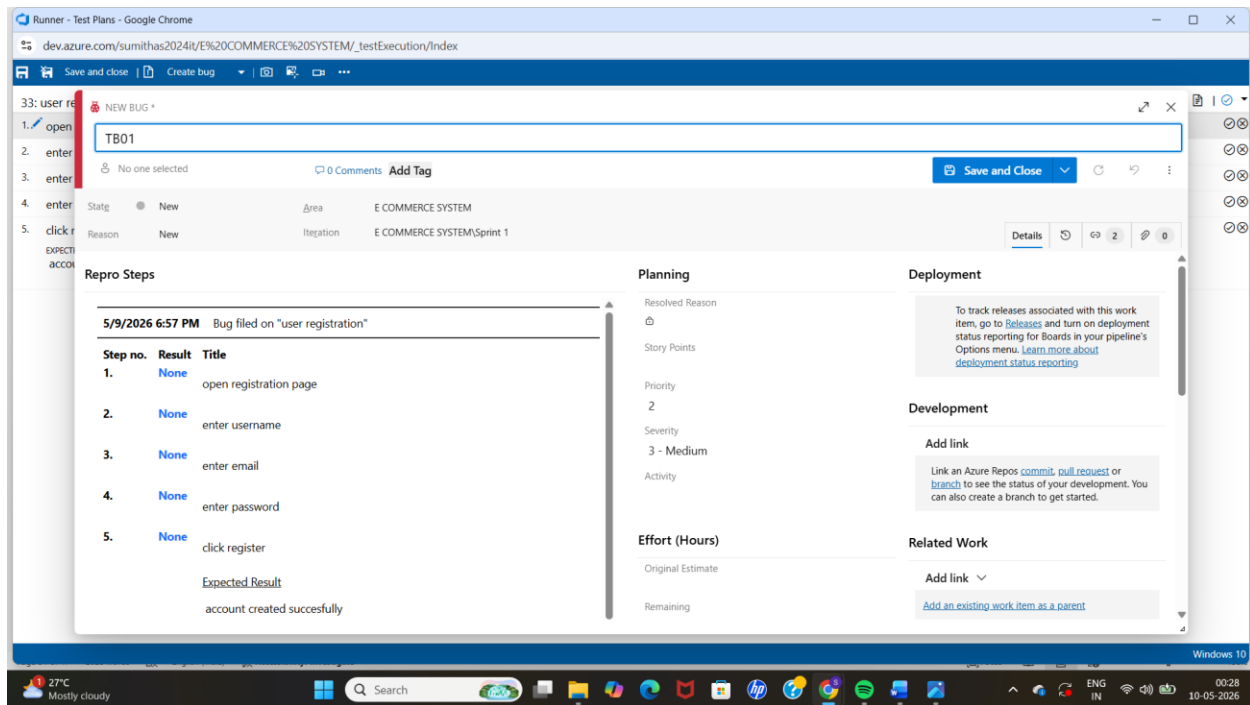
Windows 10

6. Recording the test case

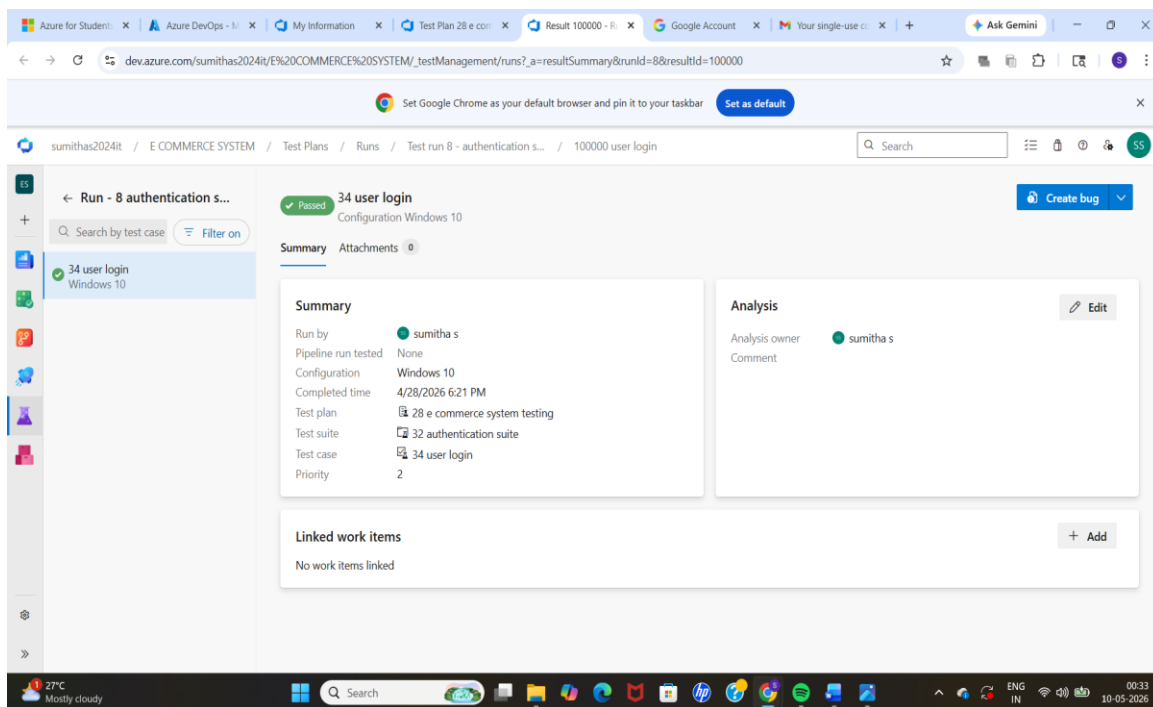


7.Creating the bug

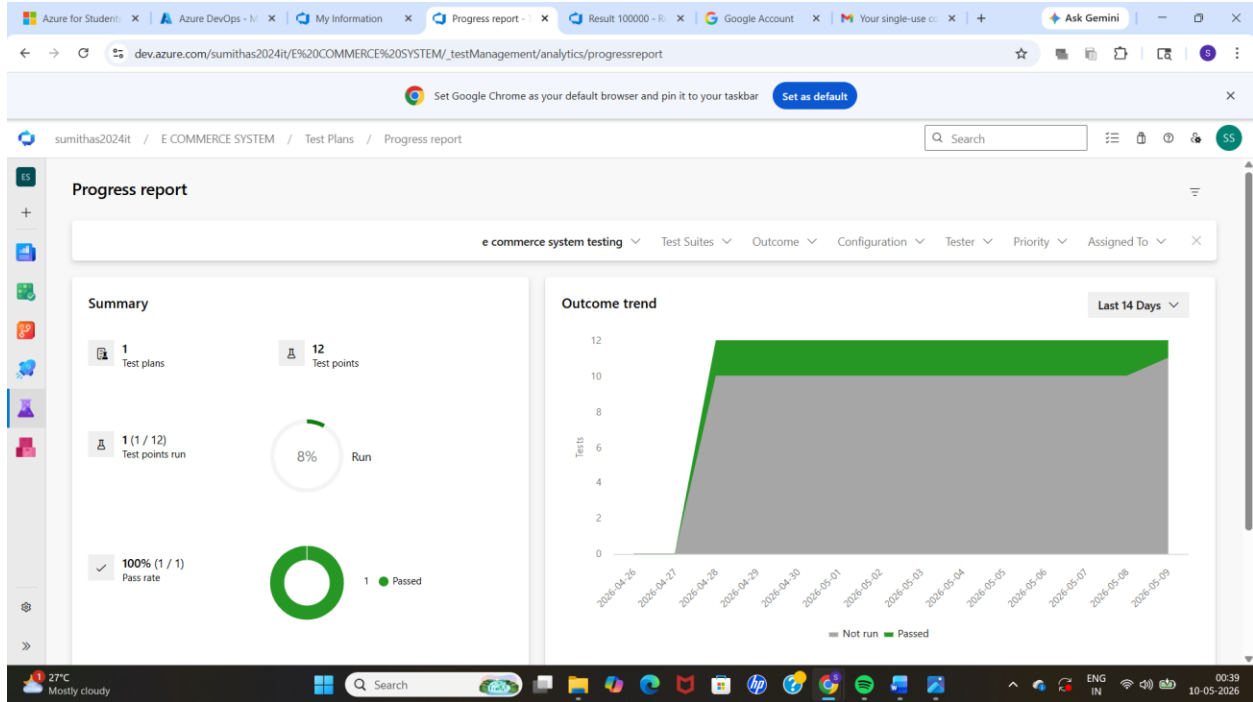




8. Test case results



9. Progress report



10.Changing the test template

Organization Settings

sumithas2024it

Search Settings

General

- Overview
- Projects
- Users
- Billing
- Global notifications
- Usage
- Extensions
- Microsoft Entra

Security

- Security overview
- Policies
- Permissions

Boards

- Process

Pipelines

- Agent pools

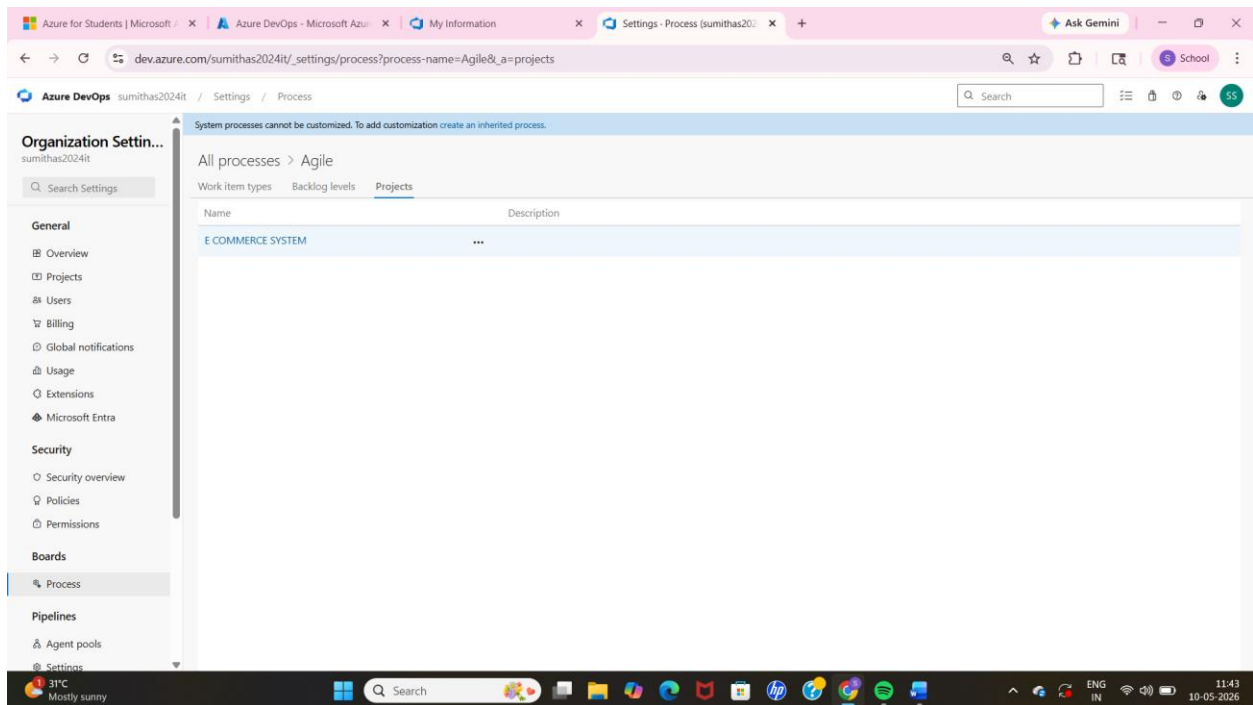
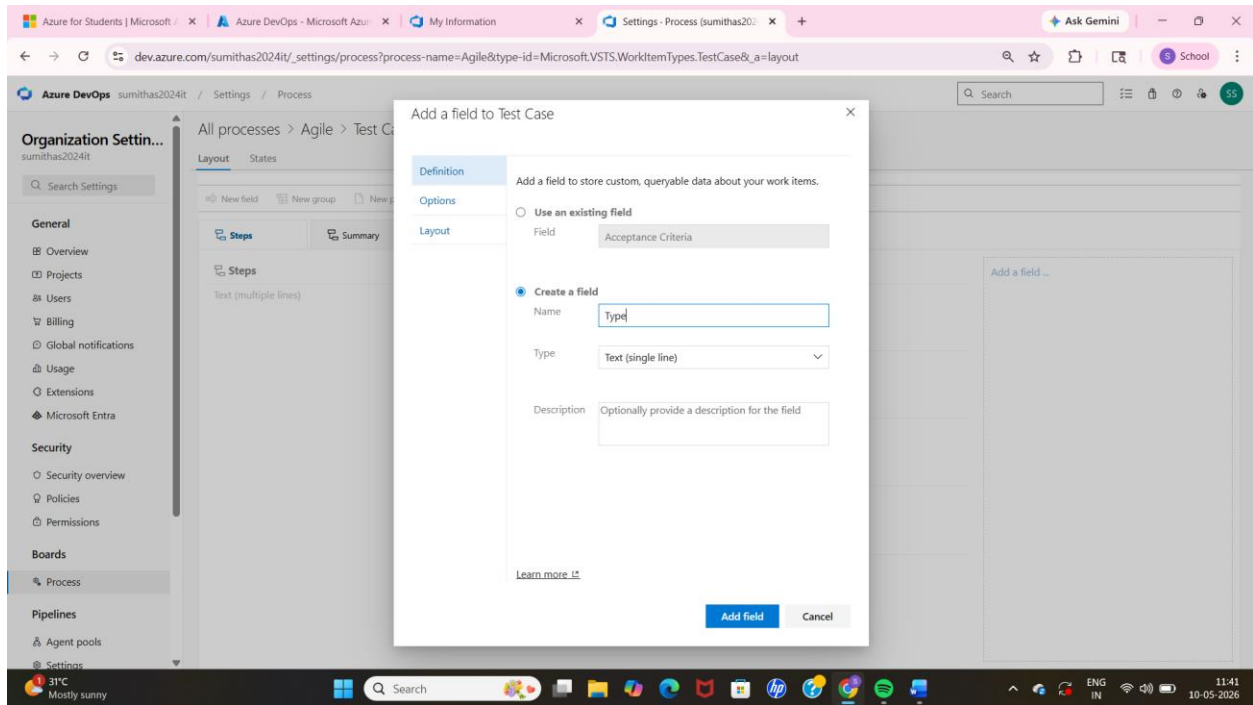
Settings

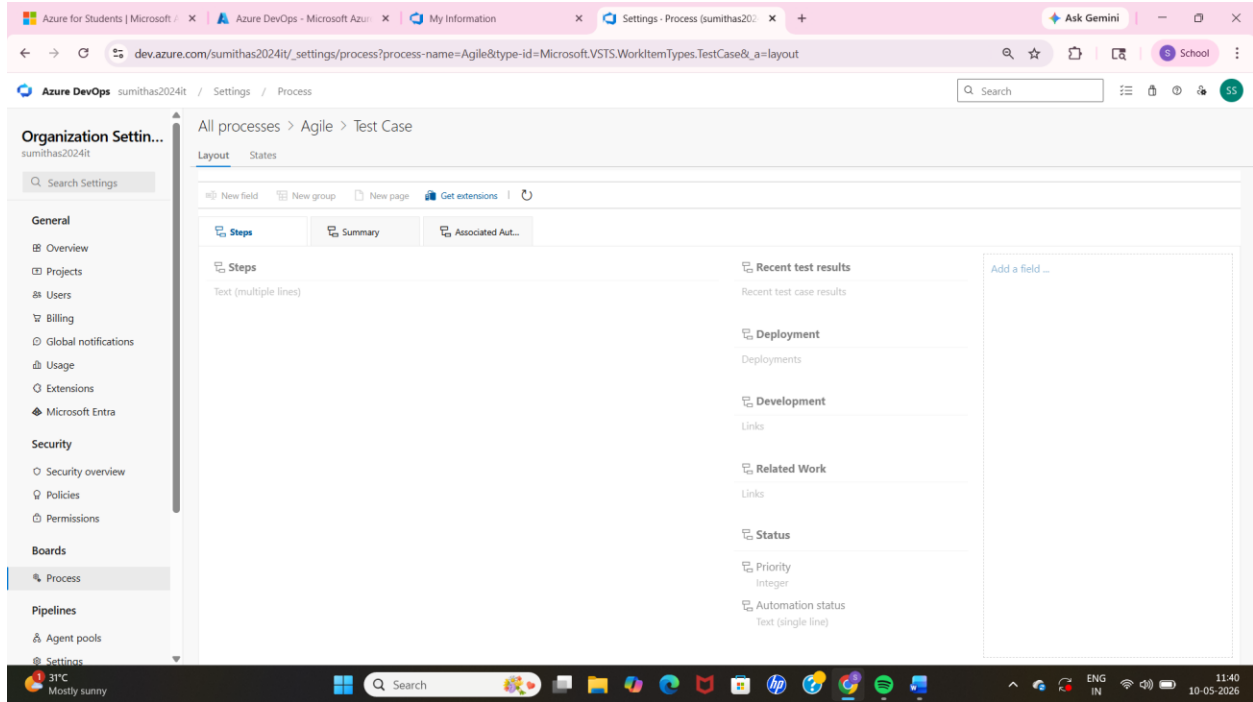
All processes

Filter by process name

Name	Description	Team projects
Basic (default)	This template is flexible for any process and great for teams getting started with Azure DevOps.	0
Agile	This template is flexible and will work great for most teams using Agile planning methods, including those practicing Scrum.	1
Scrum	This template is for teams who follow the Scrum framework.	0
CMMI	This template is for more formal projects requiring a framework for process improvement and an auditable record.	0

11. View the new test case template





Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

	LOAD TESTING AND PIPELINES
--	-------------------------------------

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Azure Load Testing:

Azure Load Testing allows you to simulate high traffic and stress tests for your web applications and APIs to understand how they perform under load. It helps identify performance bottlenecks, scalability issues, and optimize resource usage before deployment.

Steps to Create an Azure Load Testing Resource:

Before you run your first test, you need to create the Azure Load Testing resource:

1. Sign in to Azure Portal
Go to <https://portal.azure.com> and log in.
2. Create the Resource
 - Go to *Create a resource* → Search for “Azure Load Testing”.
 - Select Azure Load Testing and click Create.
3. Fill in the Configuration Details
 - *Subscription*: Choose your Azure subscription.
 - *Resource Group*: Create new or select an existing one.
 - *Name*: Provide a unique name (no special characters).
 - *Location*: Choose the region for hosting the resource.
4. (Optional) Configure tags for categorization and billing.
5. Click Review + Create, then Create.
6. Once deployment is complete, click Go to resource.

Steps to Create and Run a Load Test:

Once your resource is ready:

1. Go to your Azure Load Testing resource and click Add HTTP requests > Create.

2. Basics Tab

- *Test Name:* Provide a unique name.
- *Description:* (Optional) Add test purpose.
- *Run After Creation:* Keep checked.

3. Load Settings

- *Test URL:* Enter the target endpoint (e.g., <https://yourapi.com/products>).

4. Click Review + Create → Create to start the test.

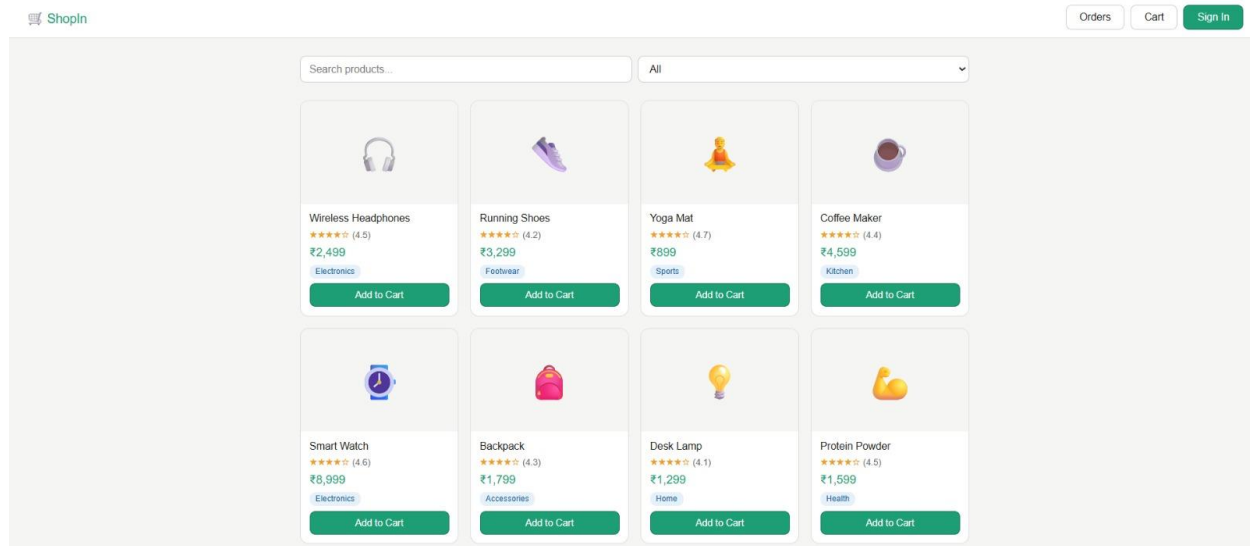
Load Testing

The screenshot displays the 'Test run 15 - product suite (Manual)' in the Azure Test Plans interface. The status is 'Completed' with a '100.00% pass rate' and '1 Passed'. The manual test is 'Test plan 28 - e commerce system testing', owned by 'sumitha s'. It was completed on 5/10/2026 at 6:16 AM, taking 19s. The pipeline run tested is 'None'. The release & release stage is also 'None'. A comment field is available for editing. Below the summary, a table shows the test results.

Outcome	Test case	Run by	Priority	Configuration	Owner	Test suite	Iteration	Area	Duration	Bugs
Passed	38 search product	sumitha s	2	Windows 10	sumitha s	35 product suite	...SYSTEM/Sprint 1	...MERCER SYSTEM	18s	0

The screenshot displays the 'Test run 14 - authentication suite (Manual)' in the Azure Test Plans interface. The status is 'Completed' with a '100.00% pass rate' and '1 Passed'. The manual test is 'Test plan 28 - e commerce system testing', owned by 'sumitha s'. It was completed on 5/10/2026 at 6:14 AM, taking 20s. The pipeline run tested is 'None'. The release & release stage is also 'None'. A comment field is available for editing. Below the summary, a table shows the test results.

Outcome	Test case	Run by	Priority	Configuration	Owner	Test suite	Iteration	Area	Duration	Bugs
Passed	44 invalid login	sumitha s	2	Windows 10	sumitha s	32 authenticatio...	...SYSTEM/Sprint 1	...MERCER SYSTEM	19s	0



Pipelines

Description:

This experiment demonstrates how to connect a GitHub-hosted Flask-based music recommendation project with Azure DevOps. The pipeline will automatically install dependencies, run basic tests, and publish artifacts. This ensures that every commit triggers checks for reliability and smooth deployment.

Steps:

1. Connect GitHub to Azure DevOps:
 - In Azure DevOps, create a new project.
 - Create a pipeline and select GitHub as the source.
 - Authorize access to your GitHub repository, ensuring that Azure DevOps can pull the repository for your pipeline.
2. Create azure-pipelines.yml in Your Repo Root:
 - In your GitHub repository, create a new file called azure-pipelines.yml in the root directory.
 - Add the following basic pipeline configuration for Python and Flask:

yml Code

trigger:

- main # Trigger pipeline when changes are pushed to the main branch

pool:

vmImage: ubuntu-latest # Use a hosted Ubuntu agent

steps:

Step 1: Checkout the code from GitHub

- checkout: self

Step 2: Set up Python environment

- task: UsePythonVersion@0

inputs:

versionSpec: '3.x' # Use the latest Python 3.x version

displayName: "Set up Python"

Step 3: Install dependencies from the correct path

- script: |

python -m pip install --upgrade pip

pip install -r project/requirements.txt # Adjusted path to requirements.txt

displayName: "Install dependencies"

Step 4: Run a simple Python script to check the environment

- script: |

python -c "print(' Hello from E-commerce!')"

displayName: "Run a Python script"

3. Pipeline Tasks Include:

- Setting up the Python environment using the UsePythonVersion task.
- Installing project dependencies from project/requirements.txt. Make sure the path to requirements.txt is correct (it is located under the project folder).
- Running a simple Python script to verify that Python is set up correctly and the pipeline works.

4. Run and Monitor Pipeline:

- Commit changes to the main branch of your repository to trigger the pipeline in Azure DevOps.
- Monitor the logs in the Azure DevOps portal to view logs, errors, or success messages and ensure everything runs smoothly.

Pipeline

The screenshot displays the Azure DevOps Pipeline interface for a project named 'E-COMMERCE SYSTEM'. The breadcrumb trail indicates the path: sumithas2024it / E-COMMERCE SYSTEM / Pipelines / ecommerce repo / 20260510.5. The pipeline run is titled '#20260510.5 • Updated App.jsx' and is marked as successful with a green checkmark. A message states: 'This run is being retained as one of 3 recent runs by main (Branch)'. Below this, the 'Summary' tab is active, showing details for an 'Individual CI by sumitha s'. The repository is 'ecommerce repo', the branch is 'main', and the commit is '26e88efd'. The build was completed 'Just now' with a duration of '17s'. It shows '0 work items' and '0 artifacts'. A 'View 34 changes' button is available. The 'Jobs' tab is also visible, showing a single job named 'Job' that completed successfully in '12s'. The left sidebar contains navigation icons for Home, Pipelines, Builds, Repositories, and Test.

sumithas2024it / E-COMMERCE SYSTEM / Pipelines / ecommerce repo / 20260510.5

Search

#20260510.5 • Updated App.jsx

ecommerce repo

Run new

This run is being retained as one of 3 recent runs by main (Branch).

View retention leases

Summary Code Coverage

Individual CI by sumitha s

Repository and version
ecommerce repo
main 26e88efd

Time started and elapsed
Just now
17s

Related
0 work items
0 artifacts

Tests and coverage
Get started

View 34 changes Parameters

Stages Jobs

Name	Status	Duration
Job	Success	12s

Result:

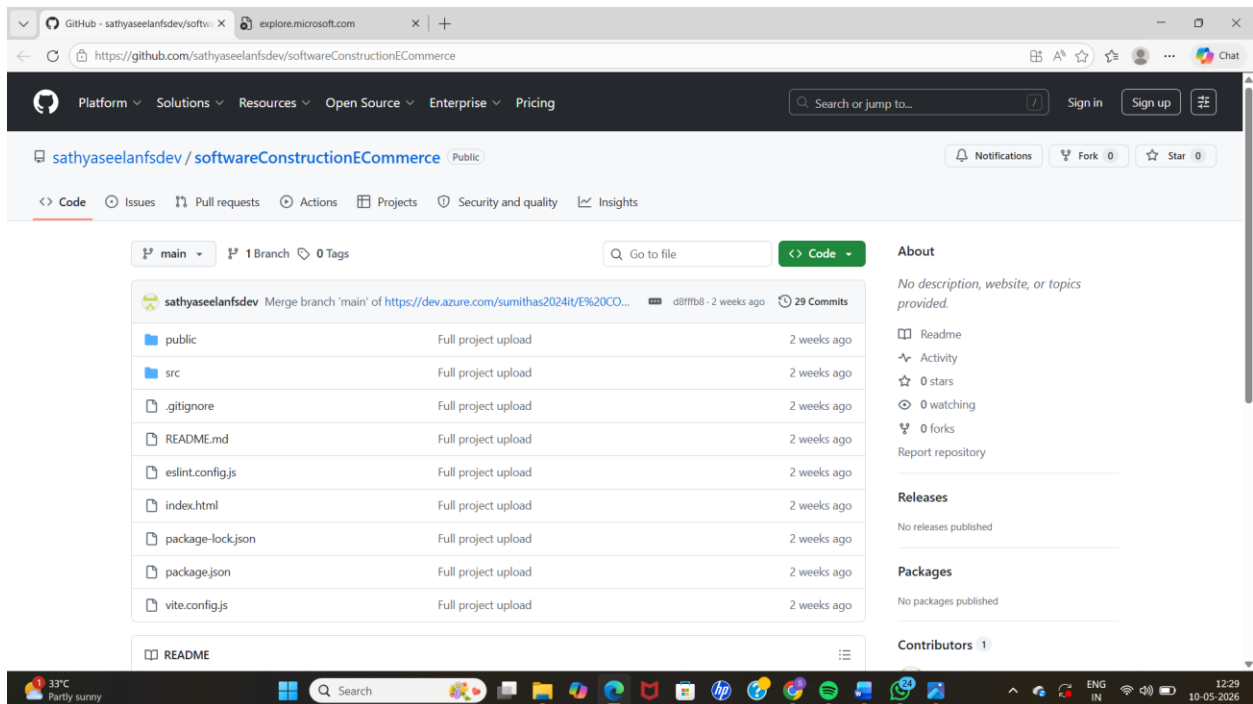
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the E-commerce project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.